Writing KVL equation we have

$$-10\text{ V} + 0.7\text{ V} + I_D(5\text{ k}\Omega) = 0$$

$$I_D = \frac{9.3\text{ V}}{5\text{ k}\Omega} = 1.86\text{ mA}$$

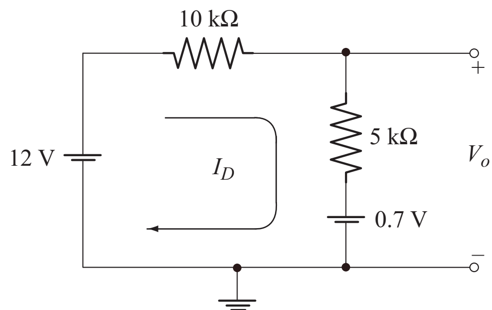
Rewriting KVL equation using V_o we have

$$-10\text{ V} + 0.7\text{ V} - V_o = 0$$

$$V_o = -9.3\text{ V}$$

To find I_D and V_o for the circuit of Fig. B

The diode is in the on state. The equivalent circuit is given below.



Writing KVL equation we have

$$12\text{ V} - I_D[10\text{ k}\Omega + 5\text{ k}\Omega] - 0.7\text{ V} = 0$$

$$I_D = \frac{11.3\text{ V}}{15\text{ k}\Omega} = 0.753\text{ mA}$$

Rewriting KVL equation using V_o we have

$$12\text{ V} - I_D(10\text{ k}\Omega) - V_o = 0$$

$$V_o = 12\text{ V} - I_D(10\text{ k}\Omega) = 4.47\text{ V}$$

Alternate method

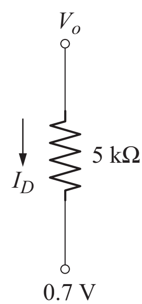


Fig. C

From Fig. C

$$I_D = \frac{V_o - 0.7\text{ V}}{5\text{ k}\Omega}$$

$$\Rightarrow \quad V_o = I_D\,(5\text{ k}\Omega) + 0.7\text{ V}$$

$$= 4.47\text{ V}$$

Example 1.13

For the circuits shown determine V_o and I_D.

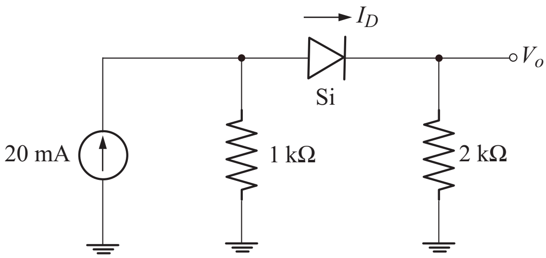


Fig. A

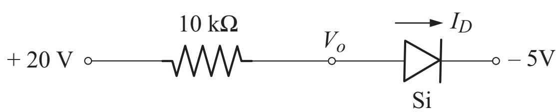
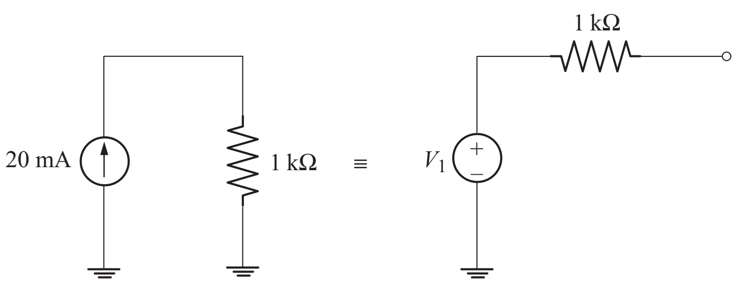


Fig. B

Solution

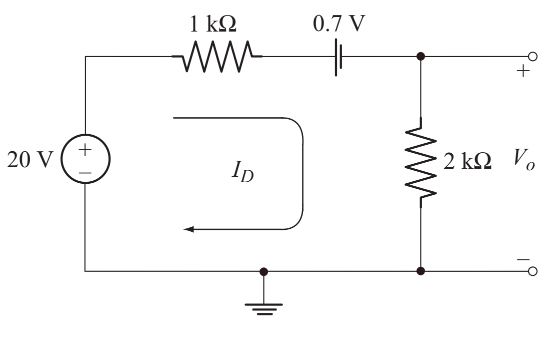
To find I_D and V_o for the circuit of Fig. A

Let us convert the current source into its equivalent voltage source.



$$V_1 = (20\text{ mA})\,(1\text{ k}\Omega) = 20\text{ V}$$

The resulting circuit is shown below



Applying KVL we get

$$20\text{ V} - I_D(1\text{ k}\Omega) - 0.7\text{ V} - I_D(2\text{ k}\Omega) = 0$$

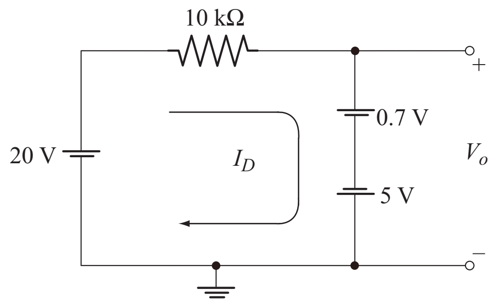
$$I_D = \frac{19.3\text{ V}}{3\text{ k}\Omega} = 6.43\text{ mA}$$

$$V_o = I_D(2\text{ k}\Omega)$$

$$= (6.43\text{ mA})(2\text{ k}\Omega) = 12.86\text{ V}$$

To find V_o and I_D for the circuit of Fig. B

The diode is in the conducting state. The equivalent circuit is shown below.



Writing KVL equation, we have

$$20\text{ V} - I_D(10\text{ k}\Omega) - 0.7\text{ V} + 5\text{ V} = 0$$

$$I_D = \frac{24.3\text{ V}}{10\text{ k}\Omega} = 2.43\text{ mA}$$

Again writing KVL equation using V_o, we have

$$20\text{ V} - I_D(10\text{ k}\Omega) - V_o = 0$$

$$V_o = 20\text{ V} - (2.43\text{ mA})(10\text{ k}\Omega) = -4.3\text{ V}$$

Example 1.14

For the circuits shown below determine I_D, V_{o_1} and V_{o_2}.

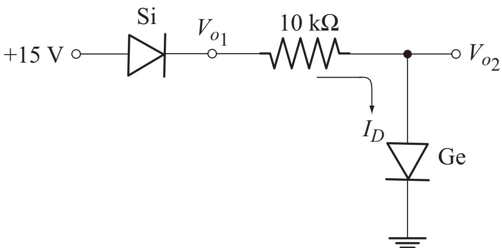


Fig. A

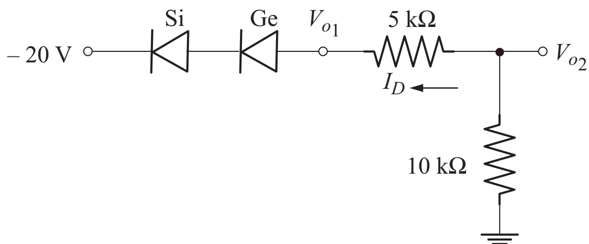
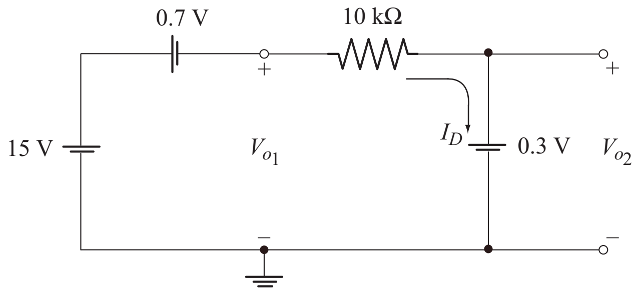


Fig. B

Solution

I_D, V_{o_1} and V_{o_2} for the circuit of Fig. A

Both diodes are in the conducting state. The equivalent circuit is shown below.



Applying KVL we have

$$15\text{ V} - 0.7\text{ V} - I_D\,(10\text{ k}\Omega) - 0.3\text{ V} = 0$$

$$I_D = \frac{14\text{ V}}{10\text{ k}\Omega} = 1.4\text{ mA}$$

$$V_{o_2} = 0.3\text{ V}$$

Writing KVL equation to the left part of the circuit we get

$$15\text{ V} - 0.7\text{ V} - V_{o_1} = 0$$

$$V_{o_1} = 14.3\text{ V}$$

Alternatively,

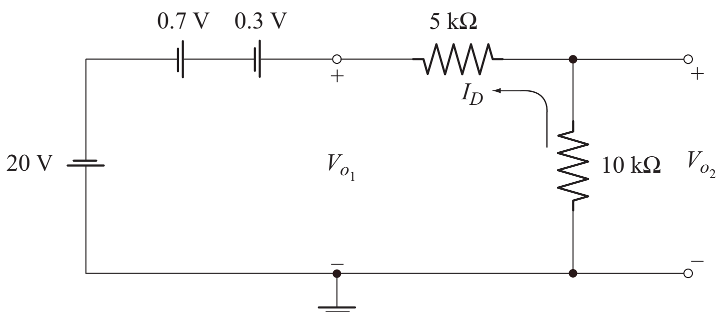
$$I_D = \frac{V_{o_1} - V_{o_2}}{10\text{ k}\Omega}$$

$$\Rightarrow \qquad V_{o_1} = I_D\,(10\text{ k}\Omega) + V_{o_2}$$

$$= (1.4\text{ mA})\,(10\text{ k}\Omega) + 0.3\text{ V} = 14.3\text{ V}$$

I_D, V_{o_1} and V_{o_2} for the circuit of Fig. B

Both diodes are in the conducting state. The equivalent circuit is shown below.



KVL equation to the circuit is

$$-20\text{ V} + 0.7\text{ V} + 0.3\text{ V} + I_D\,[5\text{ k}\Omega + 10\text{ k}\Omega] = 0$$

$$I_D = \frac{19\text{ V}}{15\text{ k}\Omega} = 1.26\text{ mA}$$

KVL equation to the left part of the circuit is

$$-20\text{ V} + 0.7\text{ V} + 0.3\text{ V} - V_{o_1} = 0$$

$$V_{o_1} = -19\text{ V}$$

$$V_{o_2} = -I_D\,(10\text{ k}\Omega) = -(1.26\text{ mA})\,(10\text{ k}\Omega) = -12.6\text{ V}$$

Example 1.15

Determine I_D, I_o, and V_o for the circuits shown below.

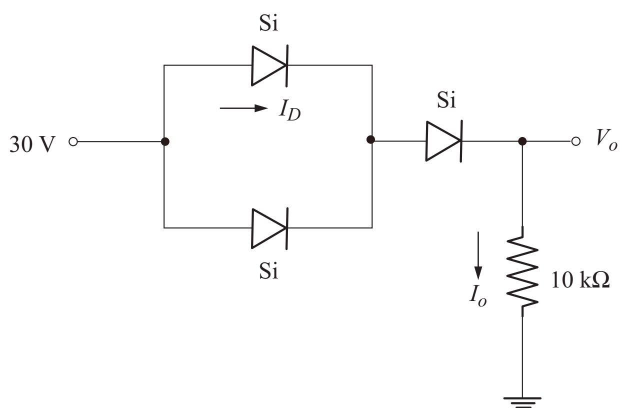


Fig. A

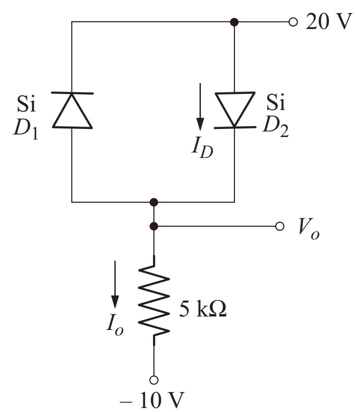
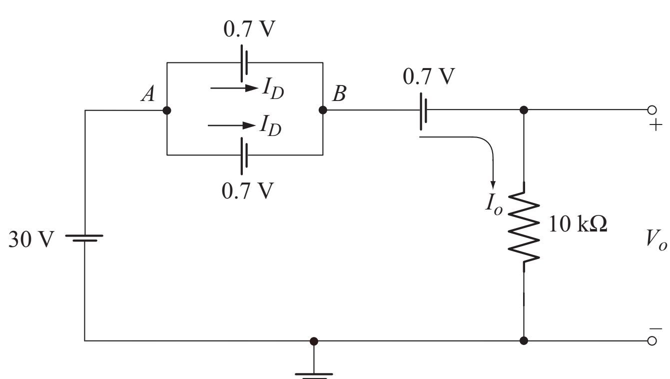


Fig. B

Solution

I_D, I_o and V_o for the circuit of Fig. A

All diodes are in the conducting state. The equivalent circuit is shown below.



Voltage between points A and B is 0.7 V.

Also $$I_o = 2I_D$$

KVL equation to the circuit is

$$30\text{ V} - 0.7\text{ V} - 0.7\text{ V} - I_o\,(10\text{ k}\Omega) = 0$$

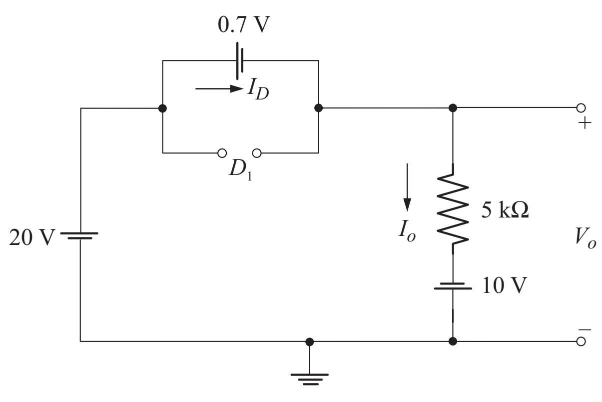
$$I_o = \frac{28.6\text{ V}}{10\text{ k}\Omega} = 2.86\text{ mA}$$

$$I_D = \frac{I_o}{2} = 1.43\text{ mA}$$

$$\begin{aligned} V_o &= I_o\,(10\text{ k}\Omega) \\ &= (2.86\text{ mA})\,(10\text{ k}\Omega) \\ &= 28.6\text{ V} \end{aligned}$$

I_D, I_o and V_D for the circuit of Fig. B

D_1 is off and D_2 is in the conducting state. The equivalent circuit is shown below.



Note that, $I_o = I_D$

KVL equation to the circuit is

$$20\text{ V} - 0.7\text{ V} - I_o\,(5\text{ k}\Omega) + 10\text{ V} = 0$$

$$I_o = \frac{29.3\text{ V}}{5\text{ k}\Omega} = 5.86\text{ mA}$$

Rewriting KVL equation using V_o,

$$20\text{ V} - 0.7\text{ V} - V_o = 0$$

$$V_o = 19.3\text{ V}$$

Alternatively: [Refer Fig. C]

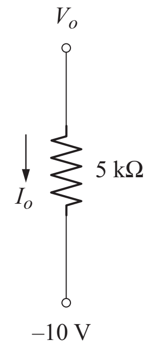


Fig. C

$$I_o = \frac{V_o - (-10\text{ V})}{5\text{ k}\Omega}$$

$$\Rightarrow \quad V_o = I_o(5\text{ k}\Omega) - 10\text{ V}$$

$$= 5.86\text{ mA }(5\text{ k}\Omega) - 10\text{ V} = 19.3\text{ V}$$

Example 1.16

For the circuits shown below calculate V_o and I.

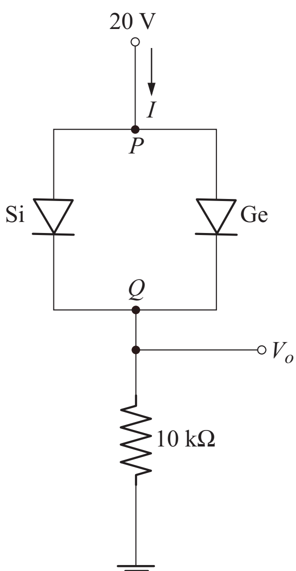


Fig. A

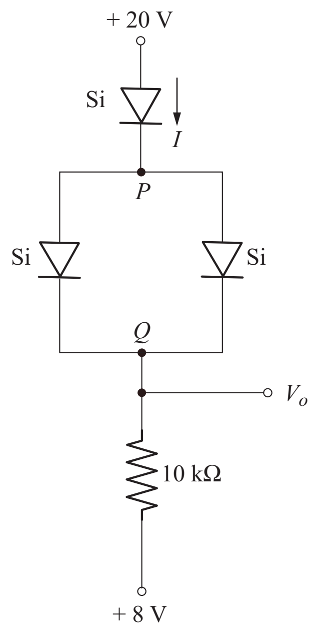
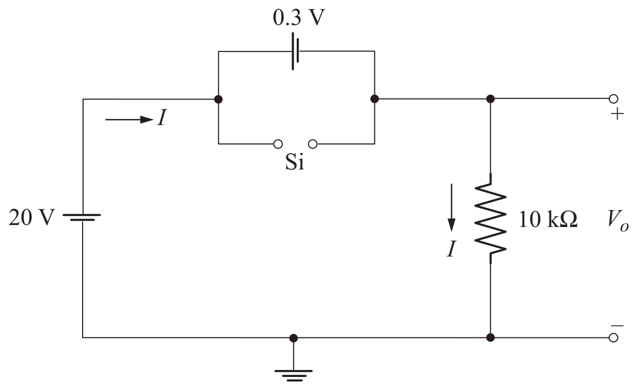


Fig. B

Solution

***I* and V_o for the Circuit of Fig. A**

Ge diode conducts and as a result the voltage between points *P* and *Q* is 0.3 V. This voltage is not adequate to turn on Si diode. Hence Si diode is permanently off and the Ge diode is permanently on. The equivalent circuit is shown below.



KVL equation to the circuit is

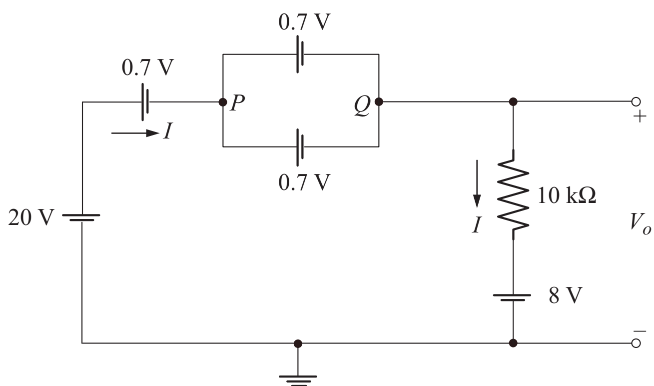
$$20\text{ V} - 0.3\text{ V} - I\,(10\text{ k}\Omega) = 0$$

$$I = \frac{19.7\text{ V}}{10\text{ k}\Omega} = 1.97\text{ mA}$$

$$V_o = I\,(10\text{ k}\Omega) = (1.97\text{ mA})\,(10\text{ k}\Omega) = 19.\,7\text{ V}$$

***I* and V_o for the Circuit of Fig. B**

All diodes are in the conducting state. The equivalent circuit is shown below.



KVL equation to the circuit is

$$20\text{ V} - 0.7\text{ V} - 0.7\text{ V} - I\,(10\text{ k}\Omega) - 8\text{ V} = 0$$

$$I = \frac{10.6\text{ V}}{10\text{ k}\Omega} = 1.06\text{ mA}$$

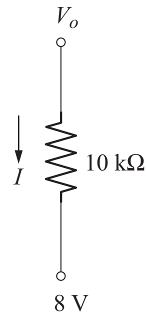


Fig. C

From Fig. C

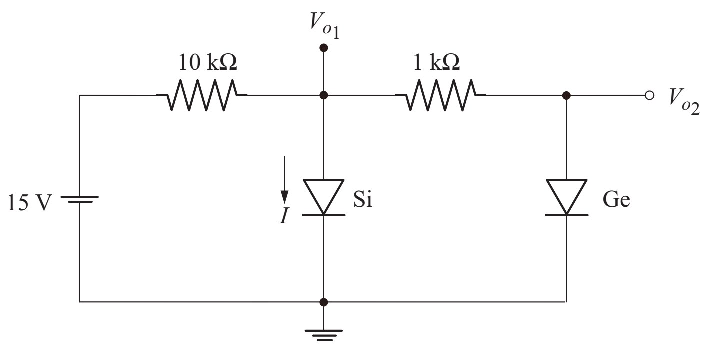
$$I = \frac{V_o - 8\text{ V}}{10\text{ k}\Omega}$$

$$\Rightarrow \quad V_o = I\,(10\text{ k}\Omega) + 8\text{ V}$$

$$= (1.06\text{ mA})\,(10\text{ k}\Omega) + 8\text{ V} = 18.6\text{ V}$$

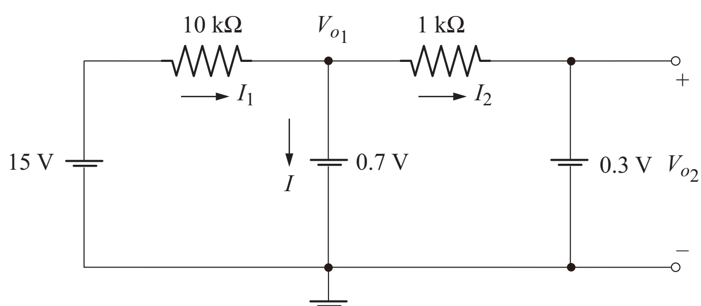
Example 1.17

For the circuit shown below determine V_{o_1}, V_{o_2} and I.



Solution

Both diodes are in the conducting state. The equivalent circuit is shown below.



From the equivalent circuit

$$V_{o_1} = 0.7 \text{ V} \quad \text{and} \quad V_{o_2} = 0.3 \text{ V}$$

KCL equation at the node V_{o_1} is

$$I_1 - I - I_2 = 0 \qquad \text{(A)}$$

$$I_1 = \frac{15 \text{ V} - V_{o_1}}{10 \text{ k}\Omega} = \frac{15 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 1.43 \text{ mA}$$

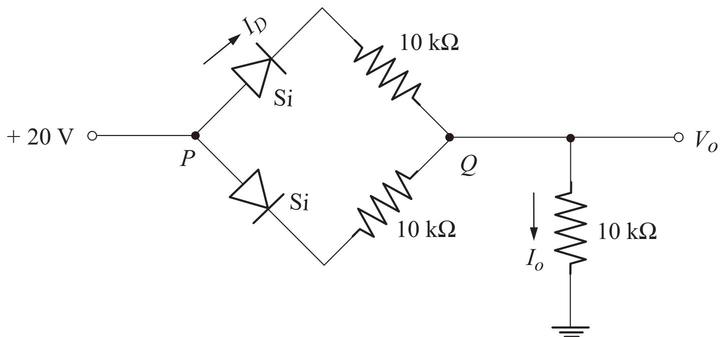
$$I_2 = \frac{V_{o_1} - V_{o_2}}{1 \text{ k}\Omega} = \frac{0.7 \text{ V} - 0.3 \text{ V}}{1 \text{ k}\Omega} = 0.4 \text{ mA}$$

Now from Equation (A),

$$I = 1.43 \text{ mA} - 0.4 \text{ mA} = 1.03 \text{ mA}$$

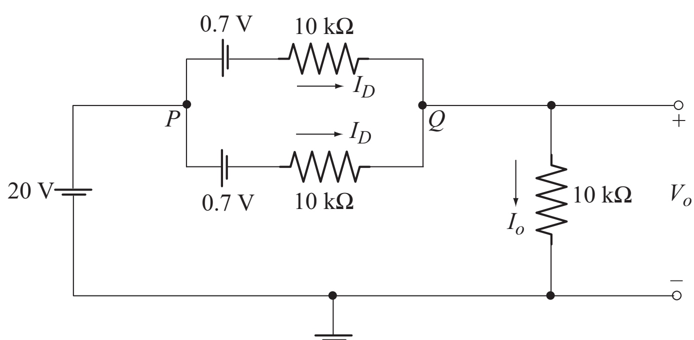
Example 1.18

For the circuit shown below calculate I_D, I_o and V_o.



Solution

Both diodes are in the conducting state. The equivalent circuit is shown below.



$$I_o = 2 I_D$$

KVL equation to the circuit is

$$20\text{ V} - 0.7\text{ V} - I_D(10\text{ k}\Omega) - I_o(10\text{ k}\Omega) = 0$$

$$I_D[10\text{ k}\Omega + 2(10\text{ k}\Omega)] = 19.3\text{ V}$$

$$I_D = 0.643\text{ mA}$$

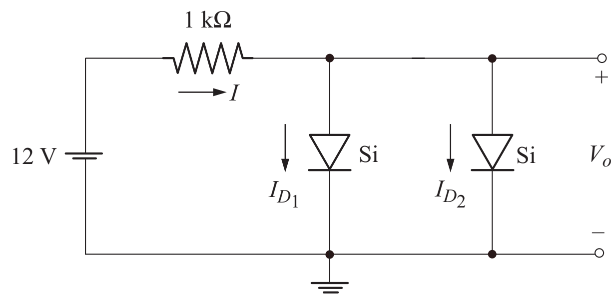
$$I_o = 2(0.643\text{ mA}) = 1.286\text{ mA}$$

$$V_o = I_o(10\text{ k}\Omega)$$

$$= (1.286\text{ mA})(10\text{ k}\Omega) = 12.86\text{ V}$$

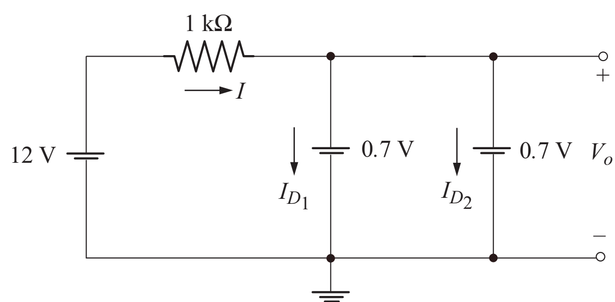
Example 1.19

For the circuit shown calculate V_o, I_{D_1}, I_{D_2} and I.



Solution

Both diodes are in the conducting state. The equivalent circuit is shown below.



From the equivalent circuit

$$V_o = 0.7\text{ V}$$

Also $$I_{D_1} = I_{D_2}$$

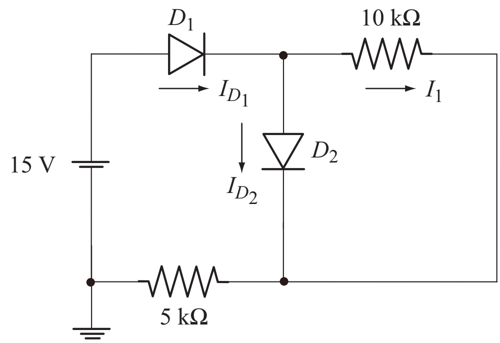
$\therefore$ $$I = 2I_{D_1} = 2I_{D_2}$$

$$I = \frac{12\text{ V} - V_o}{1\text{ k}\Omega} = \frac{12\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = 11.3\text{ mA}$$

$$I_{D_1} = I_{D_2} = \frac{I}{2}$$
$$= 5.65 \text{ mA}$$

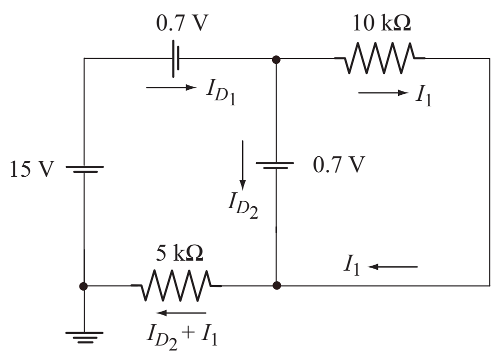
Example 1.20

For the circuit shown below calculate I_{D_1}, I_{D_2} and I_1. Both are Si diodes.



Solution

Both diodes are in the conducting state. The equivalent circuit is shown below.



$$I_{D_1} = I_1 + I_{D_2} \tag{A}$$

KVL equation to the second loop is

$$0.7 \text{ V} - I_1 (10 \text{ k}\Omega) = 0$$
$$I_1 = \frac{0.7 \text{ V}}{10 \text{ k}\Omega} = 0.07 \text{ mA}$$

KVL equation to the first loop is

$$15 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} - 5 \text{ k}\Omega\, [I_{D_2} + I_1] = 0$$
$$I_{D_2} + I_1 = \frac{13.6 \text{ V}}{5 \text{ k}\Omega} = 2.72 \text{ mA}$$

$$I_{D_2} = 2.72 \text{ mA} - I_1$$
$$= 2.72 \text{ mA} - 0.07 \text{ mA} = 2.65 \text{ mA}$$

From Equation (A)

$$I_{D_1} = 0.07 \text{ mA} + 2.65 \text{ mA} = 2.72 \text{ mA}$$

1.13 RECTIFIERS

Rectifiers convert alternating current in to direct current. Since semiconductor diodes conduct current in the forward direction and blocks current in the reverse direction they can be employed for rectification. Most of the electronic systems require a dc voltage in the range of 5 V to 30 V for the proper operation of their internal circuits. Hence it is essential to convert ac to dc.

1.14 HALF-WAVE RECTIFIER

Half-wave rectifier converters only one half cycle of ac signal in to dc. Figure 1.18 shows the circuit of half wave rectifier supplying the load *R*.

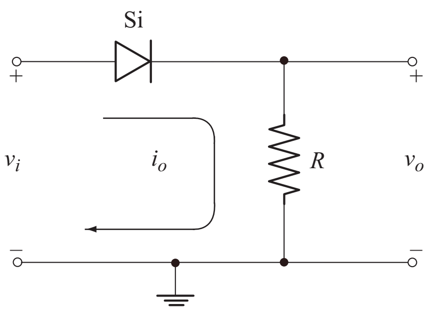


Fig. 1.18 Half-wave rectifier

v_i is the ac input voltage given by

$$v_i = V_m \sin \omega t \tag{1.24}$$

During the positive half cycle of ac input, the diode conducts when $v_i \geq V_K$. The equivalent circuit when the diode is conducting is shown in Fig. 1.19.

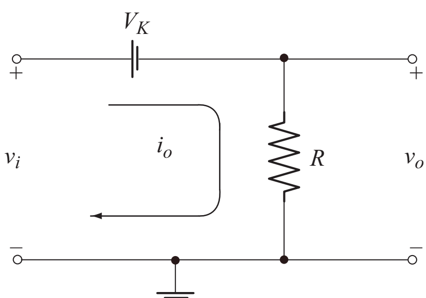


Fig. 1.19 Equivalent Circuit when the diode conducts

The KVL equation to the circuit of Fig. 1.19 is

$$v_i - V_K - v_o = 0$$

$$v_o = v_i - V_K \tag{1.25}$$

When the input signal is at its peak level V_m, the peak level of the output from Equation (1.25) is

$$V_{om} = V_m - V_K$$

Neglecting the time taken for the input signal to reach the level V_K, the output voltage when the diode is conducting can be described by the equation.

$$v_o = (V_m - V_K) \sin \omega t, \quad 0 \le \omega t \le \pi \tag{1.26}$$

The output current is given by

$$i_o = \frac{V_m - V_K}{R} \sin \omega t, \quad 0 \le \omega t \le \pi \tag{1.27}$$

When the input signal level falls below V_K, the diode stops conducting.

The equivalent circuit is shown in Fig 1.20. v_d represents the instantaneous reverse voltage across the non conducting diode.

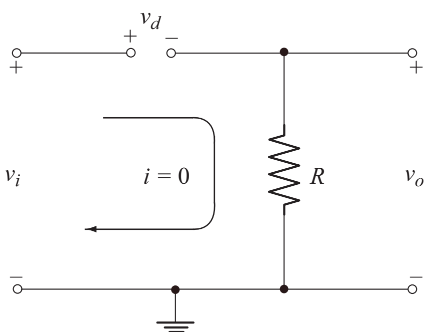


Fig. 1.20 Equivalent circuit during the negative half cycle of v_i

From the circuit of Fig 1.20, we find that

$$i_o = 0 \qquad \pi \le \omega t \le 2\pi \tag{1.28}$$

$$\therefore \quad v_o = i_o R = 0 \qquad \pi \le \omega t \le 2\pi \tag{1.29}$$

The waveforms of v_o and i_o are shown in Fig. 1.21.

Observe that both v_o and i_o are zero for $v_i \le V_K$. Since this interval is much smaller than the diode conduction time, it can be neglected.

Average (dc) Output Voltage [V_{dc}]

From Fig. 1.21 we note that the period of v_o is 2π. Hence its average or dc value is given by

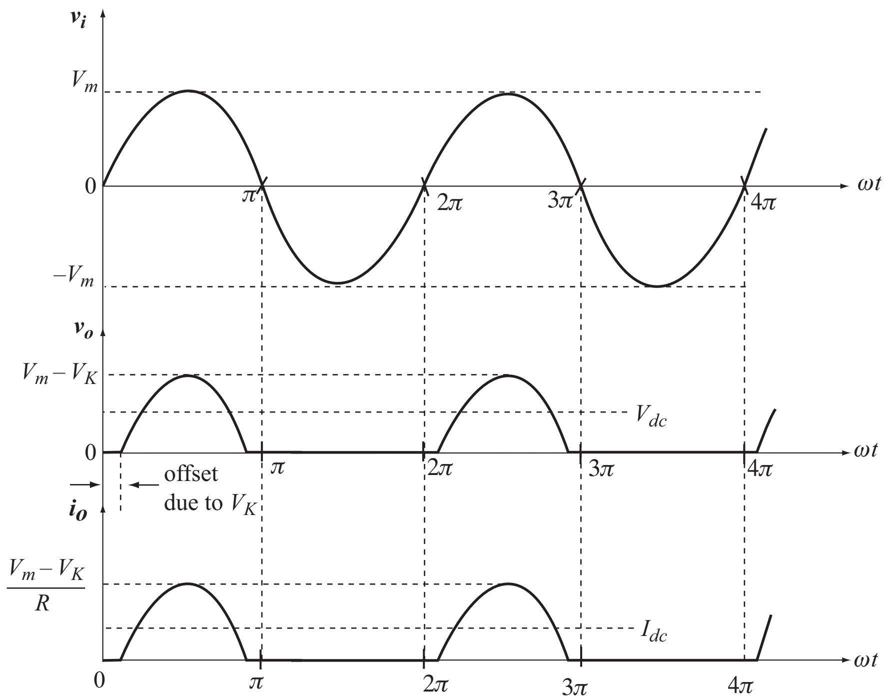


Fig. 1.21 Wave forms of v_i, v_o and i_o

$$V_{dc} = \frac{1}{2\pi}\int_0^{2\pi} v_o \, d\omega t$$

$$= \frac{1}{2\pi}\int_0^{\pi} (V_m - V_K)\sin\omega t \, d\omega t$$

$$= \frac{V_m - V_K}{2\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$V_{dc} = \frac{1}{\pi}\left[V_m - V_K\right] \tag{1.30}$$

or $$V_{dc} = 0.318\,[V_m - V_K] \tag{1.31}$$

If $V_m \gg V_K$ or if the diode is ideal, $V_K = 0$

$\therefore$ $$V_{dc} = 0.318\, V_m \tag{1.32}$$

Average (dc) Output Current [I_{dc}]

Average or dc output current is given by

$$I_{dc} = \frac{V_{dc}}{R}$$

$$I_{dc} = 0.318\,\frac{[V_m - V_K]}{R} \tag{1.33}$$

Average and Peak Diode Current

The diode current is same as the load current. Hence the average diode current equals the average load current.

i.e.,
$$I_{dc\,(diode)} = 0.318\,\frac{[V_m - V_K]}{R} \tag{1.34}$$

From Equation (1.27), the peak diode current is

$$I_{dm} = \frac{V_m - V_K}{R} \tag{1.35}$$

Note that peak diode current is same as peak load current, I_{om}.

Peak Inverse Voltage [PIV]

Diode is subjected to reverse bias during the negative half cycle of v_i. Applying KVL to the circuit of Fig 1.20 we have

$$v_i - v_d - v_o = 0$$

Since $v_o = 0$ when the diode is not conducting, we have

$$v_d = v_i$$

v_d is the instantaneous reverse voltage on the diode. Reverse voltage is maximum when $v_i = V_m$. Hence the peak inverse voltage across the diode is

$$\text{PIV} = V_m \tag{1.36}$$

For safe operation, the PIV rating of the diode must be greater than V_m.

Power Dissipation in the Diode

The power dissipated in the diode is

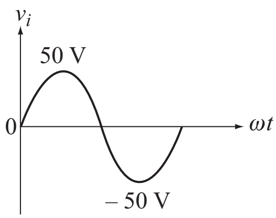
$$P_{diode} = V_K\, I_{dc\,(diode)} \tag{1.37}$$

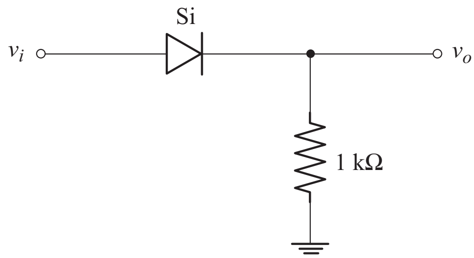
Example 1.21

For the circuit shown below calculate

(a) Average output voltage
(b) Average load current
(c) Average diode current
(d) Peak load current

(e) Peak diode current
(f) PIV of diode
(g) Power dissipation in diode





Solution

From the input voltage waveform.

$$V_m = 50 \text{ V}$$

(a) Average output voltage is

$$\begin{aligned} V_{dc} &= 0.318\,[V_m - V_K] \\ &= 0.318\,[50 \text{ V} - 0.7 \text{ V}] = 15.67 \text{ V} \end{aligned}$$

(b) Average load current is

$$\begin{aligned} I_{dc} &= \frac{V_{dc}}{R} \\ &= \frac{15.67 \text{ V}}{1 \text{ k}\Omega} = 15.67 \text{ mA} \end{aligned}$$

(c) Average diode current is

$$I_{dc\,(diode)} = I_{dc} = 15.67 \text{ mA}$$

(d) Peak load current is

$$\begin{aligned} I_{om} &= \frac{V_m - V_K}{R} \\ &= \frac{50 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = 49.3 \text{ mA} \end{aligned}$$

(e) Peak diode current is

$$I_{dm} = I_{om} = 49.3 \text{ mA}$$

(g) Peak inverse voltage is

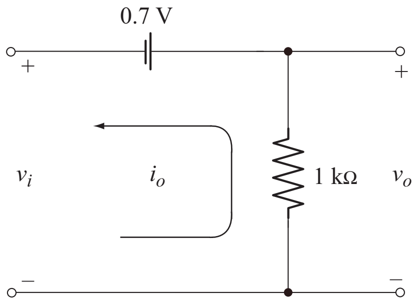
$$\text{PIV} = V_m = 50 \text{ V}$$

Example 1.22

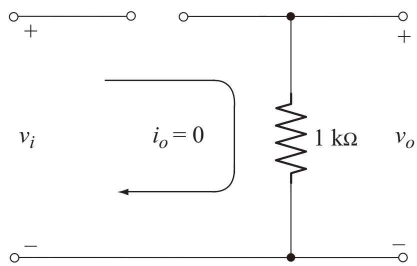
Repeat example 1.21 if the diode is reversed. Sketch the output voltage waveform.

Solution

When the diode connection is reversed, it conducts during the negative half cycle and is off during the positive half cycle. The equivalent circuits are shown below.

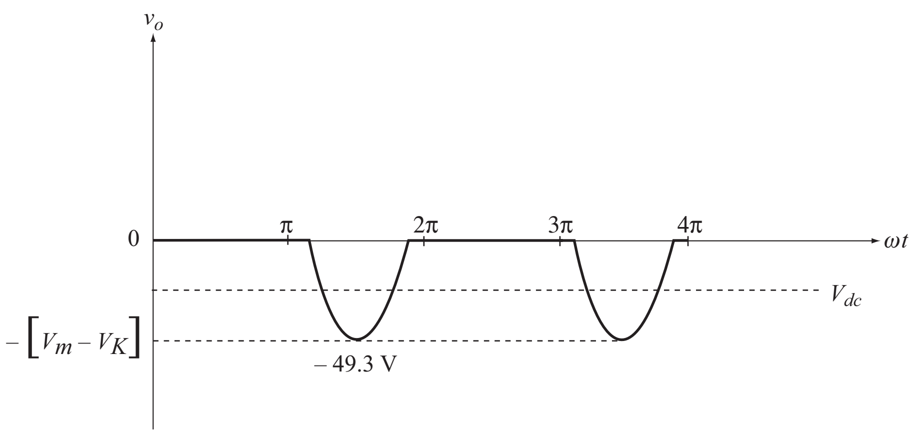


Diode on



Diode off

The output voltage waveform is shown below.



Note that the load current direction is reversed but the diode current flows from anode to cathode only. Therefore we have the following results.

$$V_{dc} = -15.67 \text{ V}$$

$$I_{dc} = -15.67 \text{ mA}$$

$$I_m = -49.3 \text{ mA}$$

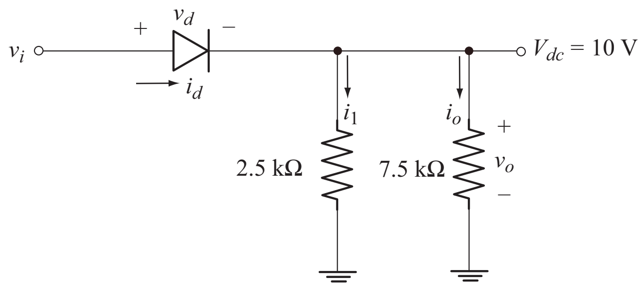
For the diode

$$I_{dc\,(\text{diode})} = 15.67 \text{ mA}$$

$$I_{dm} = 49.3 \text{ mA}$$
$$\text{PIV} = 50 \text{ V}$$

Example 1.23

For the circuit shown sketch the waveforms of i_d, v_d, v_o and i_o. Assume Si diode.



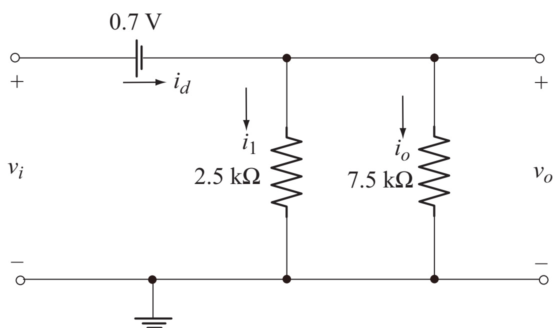
Solution

$$V_{dc} = 0.318\,[V_m - V_K]$$
$$10 \text{ V} = 0.318\,[V_m - 0.7 \text{ V}]$$

Solving we get

$$V_m = 32.15 \text{ V}$$
$$\therefore \quad v_i = V_m \sin \omega t$$
$$= 32.15 \sin \omega t \text{ Volts}$$

The equivalent circuit when the diode is conducting is shown below.



Peak value of output voltage is

$$V_{om} = V_m - V_K$$
$$= 32.15 \text{ V} - 0.7 \text{ V}$$
$$= 31.45 \text{ V}$$
$$\therefore \quad v_o = 31.45 \sin \omega t \text{ V} \qquad \text{(A)}$$

$$i_o = \frac{v_o}{7.5\ \text{k}\Omega}$$

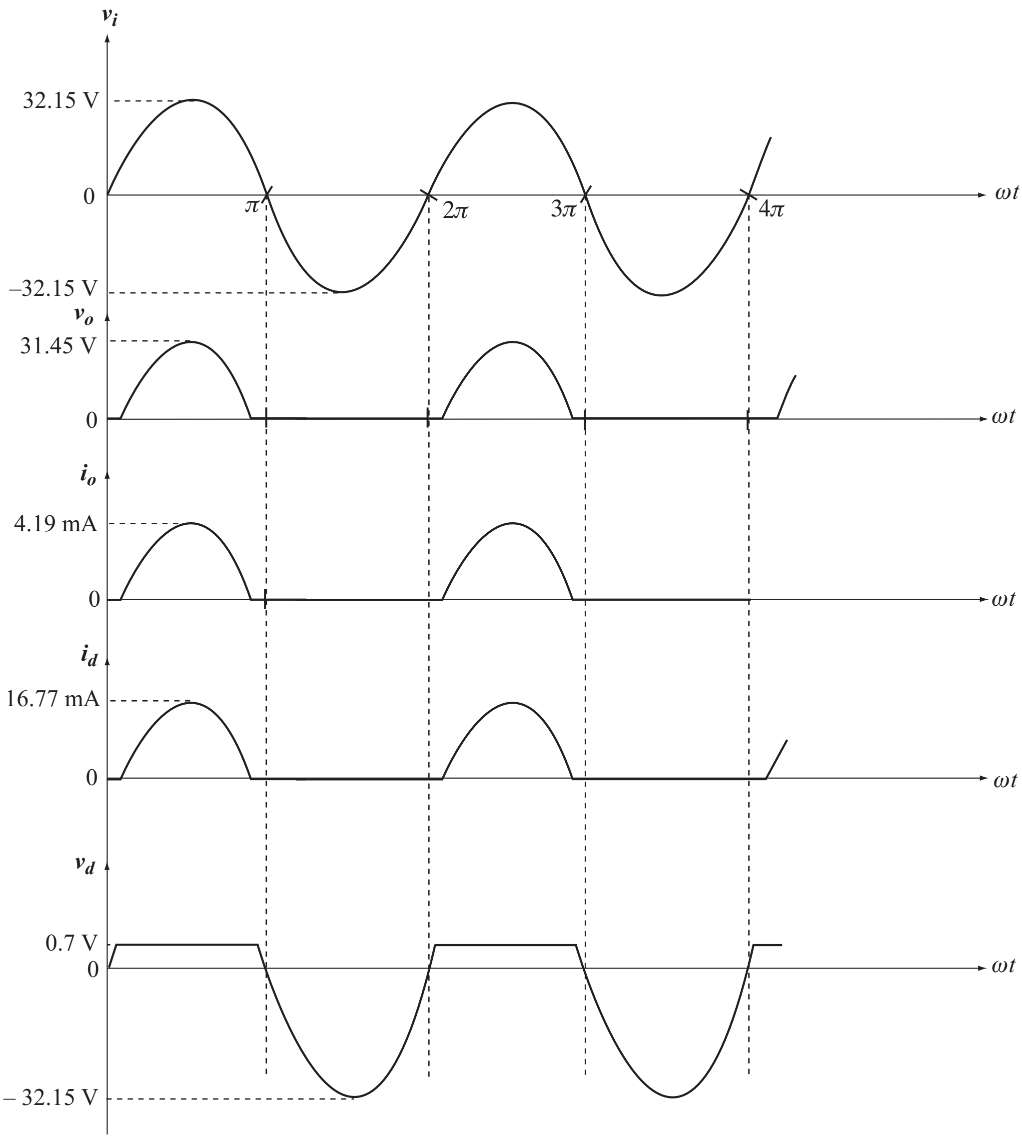
$$\therefore \quad i_o = 4.\ 19 \sin \omega t \text{ mA} \tag{B}$$

$$i_l = \frac{v_o}{2.5\ \text{k}\Omega} = 12.58 \sin \omega t \text{ mA}$$

$$i_d = i_l + i_o$$

$$= 16.\ 77 \sin \omega t \text{ mA} \tag{C}$$

The waveforms of i_d, v_d, v_o and i_o are shown below.



Note:

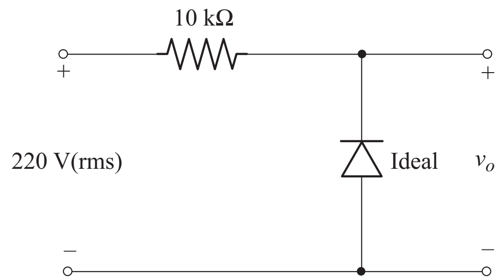
- If the diode is ideal, $V_K = 0$

$$\therefore \qquad V_{om} = V_m = 32.15 \text{ V}$$

- Without 2.5 kΩ resistor, $i_o = i_d$.

Example 1.24

For the circuit shown below sketch v_o and calculate V_{dc}.



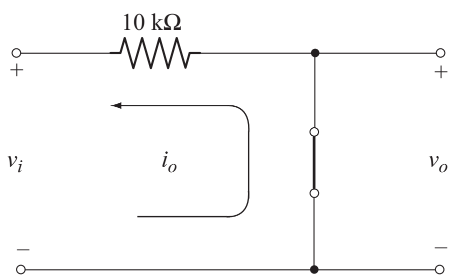
Solution

$$V_i = 220 \text{ V (rms)}$$

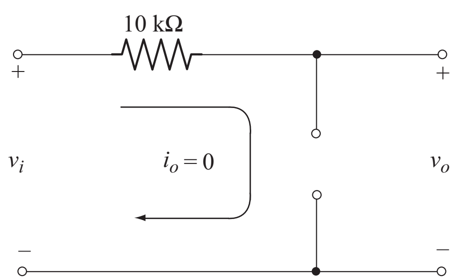
$$V_m = 220\sqrt{2} = 311.13 \text{ V}$$

Since the diode is ideal $V_K = 0$.

Diode conducts during the negative half cycle of v_i and is off during the positive half cycle. The equivalent circuits are shown below.



(a) Diode on

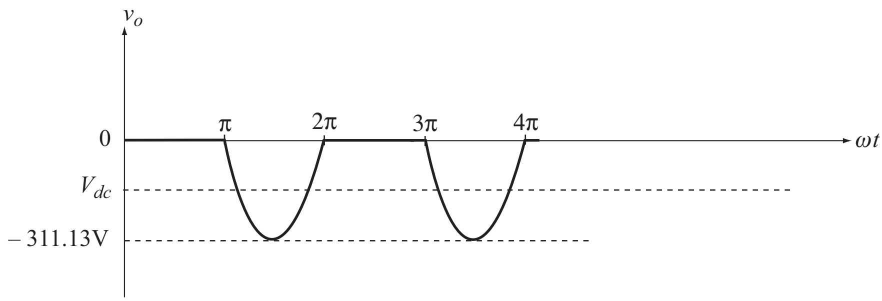


(b) Diode off

From the equivalent circuits we can write

$$v_o = \begin{cases} 0 & 0 \le \omega t \le \pi \\ v_i & \pi \le \omega t \le 2\pi \end{cases}$$

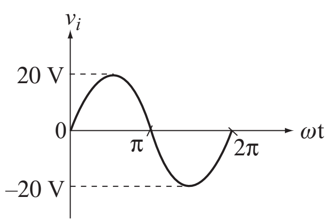
The waveform of v_o is shown below.



$$V_{dc} = -0.318\,[V_m - V_K]$$
$$= -0.318\,[311.13\text{ V}] \quad [\text{since } V_K = 0]$$
$$= -98.94\text{ V}$$

Example 1.25

For the circuit shown below sketch v_o and i_R.



Solution

The equivalent circuits are shown below.

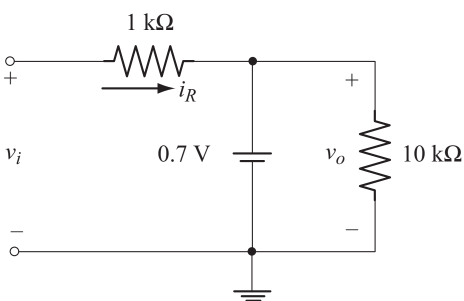


Fig. A Equivalent circuit during diode conduction

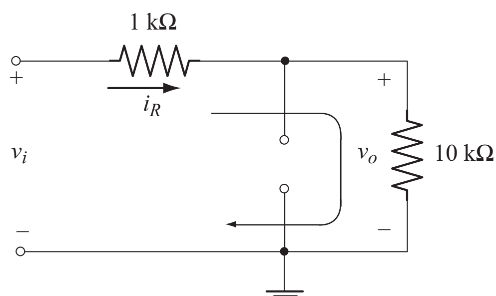


Fig. B Equivalent circuit when diode is off

***For* $0 \le \omega t \le \pi$**

Diode conducts. We refer the circuit of Fig. A

$$v_o = 0.7 \text{ V (constant)}$$

$$i_R = \frac{v_i - 0.7 \text{ V}}{1 \text{ k}\Omega}$$

When $v_i = V_m = 20$ V, the corresponding current is

$$I_{Rm} = \frac{20 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = 19.3 \text{ mA}$$

***For* $\pi \le \omega t \le 2\pi$**

Diode is off. We refer the circuit of Fig. B.

$$i_R = \frac{v_i}{1 \text{ k}\Omega + 10 \text{ k}\Omega}$$

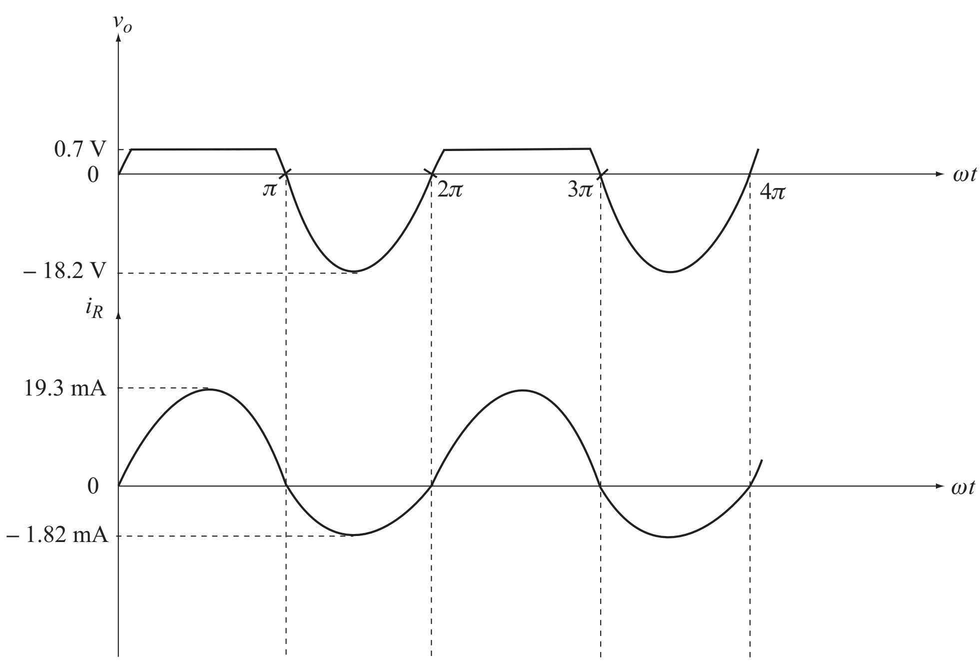
When $v_i = -20$ V

$$I_{Rm} = \frac{-20 \text{ V}}{11 \text{ k}\Omega} = -1.82 \text{ mA}$$

$$v_o = i_R (10 \text{ k}\Omega)$$

$$\Rightarrow \quad V_{om} = I_{Rm}(10 \text{ k}\Omega) = -18.2 \text{ V}$$

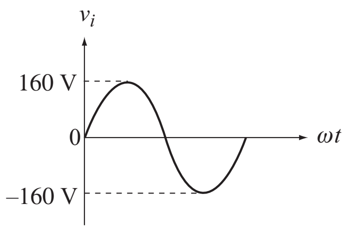
Wave forms of v_o and i_R are shown below.

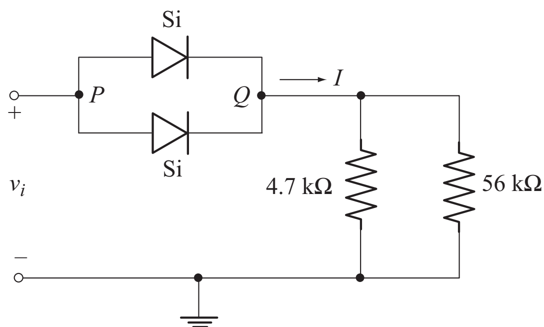


Example 1.26

For the circuit shown below determine

(a) The maximum current rating of each diode if P_{max} for each diode is 14 mW.
(b) The maximum value of I.
(c) The maximum current through each diode.
(d) Check whether the diode current rating is exceeded.
(e) If one diode is removed, calculate the maximum current through other diode. Will this current be within the maximum current rating of diode?





Solution

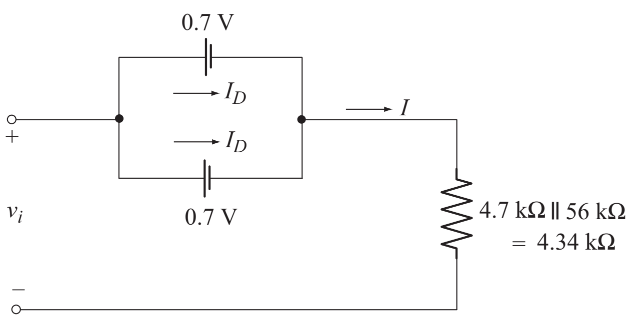
(a)

$$P_{max} = (0.7\text{ V})\, I_D$$

$$I_D = \frac{P_{max}}{0.7\text{ V}} = \frac{14\text{ mW}}{0.7\text{ V}} = 20\text{ mA}$$

Maximum forward current rating of each diode is 20 mA.

(b) Both diodes conduct during the positive half cycle of v_i. The equivalent circuit is shown below.



$$I = 2 I_D$$

KVL equation to the circuit is

$$v_i - 0.7 \text{ V} - I\,(4.34 \text{ k}\Omega) = 0$$

$$I = \frac{v_i - 0.7 \text{ V}}{4.34 \text{ k}\Omega}$$

Maximum value of I_m occurs when v_i is maximum.

$$\therefore \qquad I_{\max} = \frac{160 \text{ V} - 0.7 \text{ V}}{4.34 \text{ k}\Omega} = 36.7 \text{ mA}$$

$$I_{D\max} = \frac{I_{\max}}{2} = 18.35 \text{ mA} = I_{dm}$$

(c) Note that $I_{D\max} < 20$ mA
$\therefore$ Diode current rating is not exceeded.

(d) With only one diode

$$I = I_D$$

$$\therefore \qquad I_{\max} = I_{D\max} = 36.7 \text{ mA}$$

In this case the diode current rating is exceeded. The diode carries almost twice the rated current which is not safe.

◆ 1.15 FULL-WAVE BRIDGE RECTIFIER

Full wave rectifier converts both half cycle of ac signal in to dc. Figure 1.22 shows the circuit of full-wave bridge rectifier.

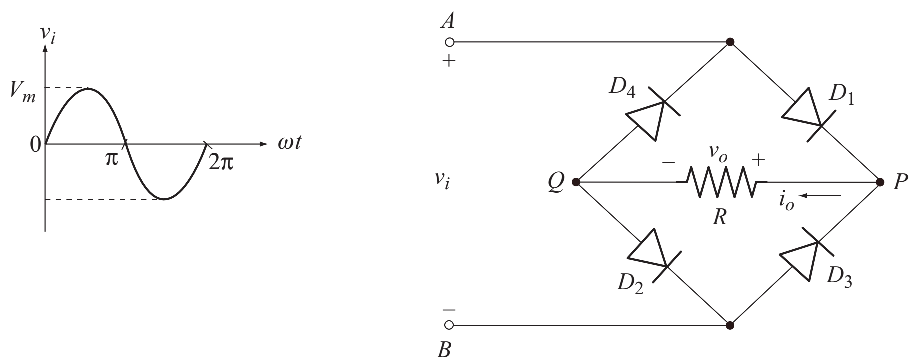


Fig. 1.22 Full-wave bridge rectifier

During the positive half cycle of v_i, diodes D_1 and D_2 are forward biased and D_3 and D_4 are reverse biased. The equivalent circuit is shown in Fig. 1.23(a). The current follows the path $A - D_1 - P - Q - D_2 - B - A$.

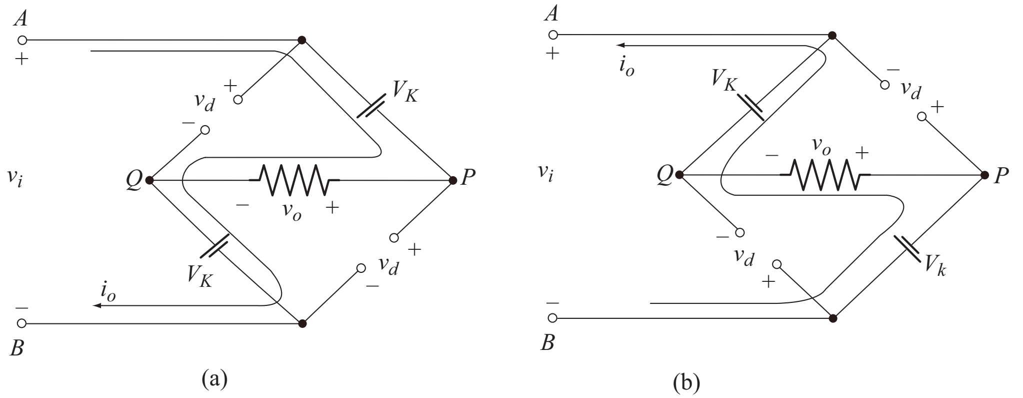


Fig. 1.23 Equivalent circuit during
(a) Positive half cycle of v_i
(b) Negative half cycle of v_i

Applying KVL to the current path indicated in Fig. 1.23 (a) we have

$$v_i - V_K - v_o - V_K = 0$$

$$v_o = v_i - 2V_K \tag{1.38}$$

When v_i is at its peak value V_m, the peak level of v_o is

$$V_{om} = V_m - 2\,V_K \tag{1.39}$$

Now the output voltage equation is

$$v_o = (V_m - 2V_K)\sin\omega t, \quad 0 \le \omega t \le \pi \tag{1.40}$$

The output current is

$$i_o = \frac{V_m - 2V_K}{R}\sin\omega t$$

or

$$i_o = I_{om}\sin\omega t, \quad 0 \le \omega t \le \pi \tag{1.41}$$

where I_{om} is the peak output / load current , given by

$$I_{om} = \frac{V_m - 2V_K}{R} \tag{1.42}$$

During the negative half cycle of v_i, diodes D_3 and D_4 conduct and D_1 and D_2 are reverse biased. The equivalent circuit is shown in Fig 1.23 (b). The current follows the path $B - D_3 - P - Q - D_4 - A - B$. Note that, during both half cycles of v_i, the load current i_o, flows from P to Q only. Hence i_o is undirectional or dc.

Equations (1.40) and (1.41) are also applicable for the interval $\pi \le \omega t \le 2\pi$. In Fig. 1.23 , v_d represents the instantaneous reverse voltage across the non conducting diodes. The waveforms of v_o and i_o are shown in Fig 1.24.

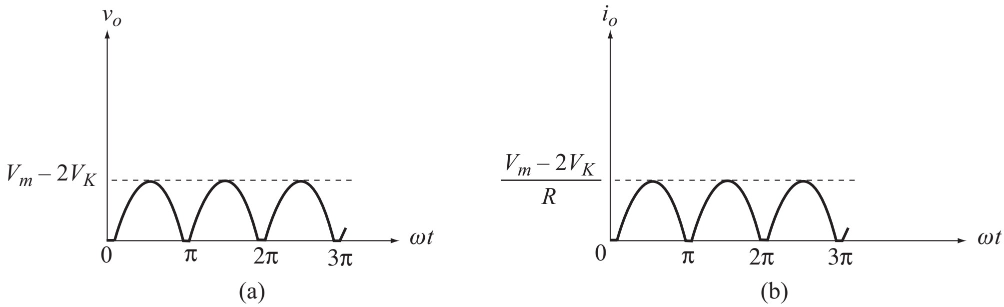


Fig. 1.24 Wave forms of v_o and i_o

Average (dc) Output Voltage [V_{dc}]

From Fig. 1.24 we note that the period of v_o is π. Hence the average or dc value is given by

$$V_{dc} = \frac{1}{\pi}\int_0^{\pi} v_o \, d\omega t$$

$$= \frac{1}{\pi}\int_0^{\pi} \left[V_m - 2V_K\right] \sin\omega t \, d\omega t$$

$$= \frac{V_m - 2V_K}{\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$V_{dc} = \frac{2\left[V_m - 2V_K\right]}{\pi}$$

or $$V_{dc} = 0.636\,[V_m - 2V_K] \tag{1.43}$$

If $V_m \gg V_K$ or if the diode is ideal, $V_K = 0$

$\therefore$ $$V_{dc} = 0.636\,V_m \tag{1.44}$$

Note that, the dc output voltage of full-wave rectifier is twice that of half wave rectifier.

Average (dc) Output Current [I_{dc}]

The average or dc output current is given by

$$I_{dc} = \frac{V_{dc}}{R}$$

$$I_{dc} = 0.636\,\frac{[V_m - 2V_K]}{R} \tag{1.45}$$

Average and Peak Diode Current

Diodes D_1 and D_2 Conducts during the positive half cycle of v_i and D_3 and D_4 during the negative half cycle. Hence the load current I_{dc} is equally shared by each pair of diode. Also the Conducting diodes are in series. Hence dc current through each diode is

$$I_{dc\ (diode)} = \frac{I_{dc}}{2}$$

$$= 0.318\ \frac{[V_m - 2\ V_K]}{R} \qquad (1.46)$$

The peak diode current is same as the peak load current. Hence from equation 1.42,

$$I_{dm} = I_{om} = \frac{V_m - 2\ V_K}{R} \qquad (1.47)$$

***Peak Inverse Voltage* [*PIV*]**

Applying KVL to the path $A - Q - B - A$ in the circuit of Fig 1.23 (a), we get

$$v_i - v_d - V_K = 0$$

$$v_d = v_i - V_K$$

v_d is the instaneous reverse voltage on the diode. Maximum reverse voltage occurs when, $v_i = V_m$. Hence peak inverse voltage across the diode is

$$\text{PIV} = V_m - V_K$$

For $V_m \gg V_K$,

$$\text{PIV} = V_m \qquad (1.48)$$

It is important to note that, the PIV rating of the diode should be greater then V_m, for safe operation.

Diode Power Dissipation

The power dissipated in each diode is

$$P_{diode} = V_K\, I_{dc\ (diode)} \qquad (1.49)$$

Example 1.27

A full-wave bridge rectifier with a 220 V rms sinusoidal input has a load resistor of 10 kΩ. Assuming silicon diodes, calculate

(a) dc output voltage and dc load current
(b) peak output voltage and peak load current
(c) peak and average diode currents
(d) power rating of each diode
(e) PIV across each diode

Solution

$$V_i = 220\text{ V (rms)}$$

$$V_m = 220\text{ V} \times \sqrt{2} = 311.13\text{ V}$$

$$R = 10\text{ k}\Omega$$

(a) DC output voltage is

$$V_{dc} = 0.636\,[V_m - 2\,V_K]$$
$$= 0.636\,[311.13\text{ V} - 1.4\text{ V}] = 197\text{ V}$$

dc load current is

$$I_{dc} = \frac{V_{dc}}{R}$$
$$= \frac{197\text{ V}}{10\text{ k}\Omega} = 19.7\text{ mA}$$

(b) Peak output voltage is

$$V_{om} = V_m - 2\,V_K$$
$$= 311.13\text{ V} - 1.4\text{ V} = 309.73\text{ V}$$

Peak load current is

$$I_{om} = \frac{V_{om}}{R}$$
$$= \frac{309.72\text{ V}}{10\text{ k}\Omega} = 30.972\text{ mA}$$

(c) Peak diode current is same as peak load current

$$\therefore \quad I_{dm} = I_{om} = 30.972\text{ mA}$$

Average diode current is

$$I_{dc\ (diode)} = \frac{I_{dc}}{2}$$
$$= \frac{19.7\text{ mA}}{2} = 9.85\text{ mA}$$

(d) Diode power dissipation is

$$P_{diode} = V_K\,I_{dc\ (diode)}$$
$$= (0.7\text{ V})\,(9.85\text{ mA}) = 6.895\text{ mW}$$

If the peak diode current is considered

$$P_{D\max} = V_K\,I_{dm}$$
$$= (0.7\text{ V})\,(30.972\text{ mA}) = 21.68\text{ mW}$$

For safe operation, it is wise to consider the maximum power dissipation $P_{D\max}$.

(e) PIV across each diode is

$$\text{PIV} = V_m$$
$$= 311.13\text{ V}$$

Example 1.28

For the circuit shown below.

(a) Explain the operation of the circuit

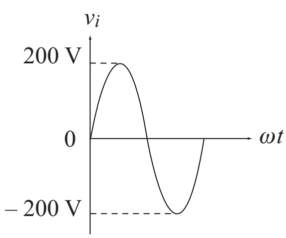
(b) Calculate the dc output voltage and current.

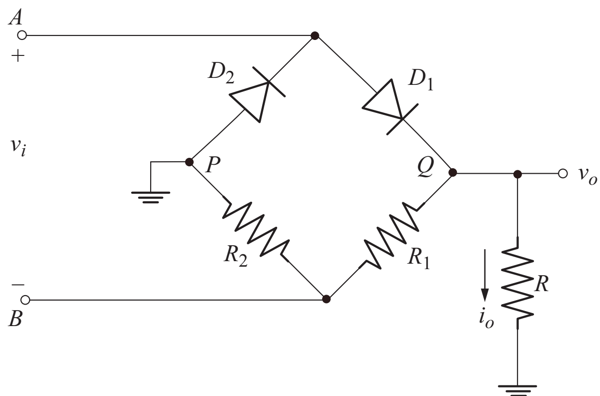
(c) Find the average and peak diode currents

(d) Calculate the required PIV rating of each diode.

Assume ideal diodes.

Take $R_1 = R_2 = R = 10$ kΩ.





Solution

Since diodes are ideal, $V_K = 0$.

(a) *Circuit Operation*

During the positive half cycle of v_i, D_1 conducts and D_2 is reverse biased. The equivalent circuit is shown below. R can be connected between P and Q.

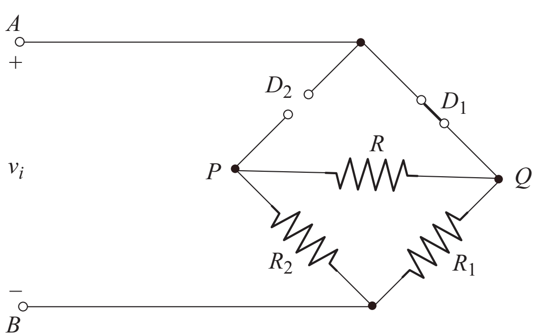


Fig. A

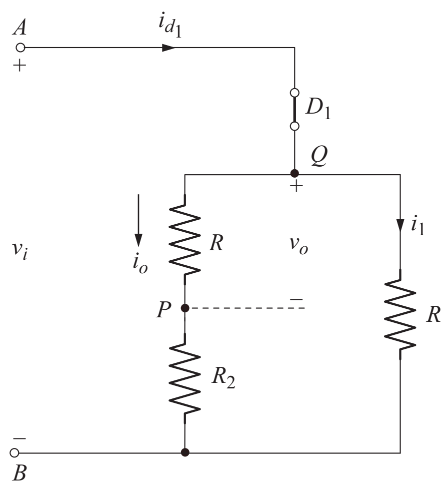


Fig. B

From Fig. B
The output current is

$$i_o = \frac{v_i}{R + R_2}$$

$$= \frac{200 \sin \omega t}{20 \text{ k}\Omega}$$

$$= 10 \sin \omega t \text{ mA}, \quad 0 \le \omega t \le \pi \qquad \text{(A)}$$

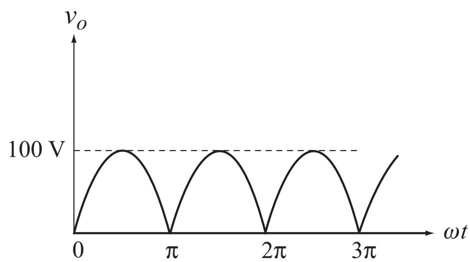
The output voltage is

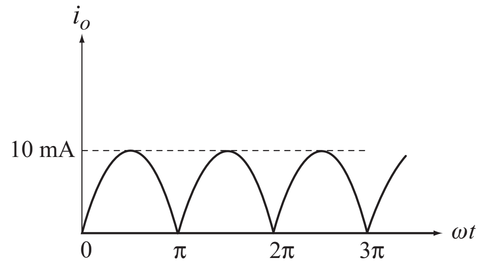
$$v_o = i_o R$$

$$= (10 \sin \omega t \text{ mA})\,(10 \text{ k}\Omega)$$

$$= 100 \sin \omega t \text{ V}, \quad 0 \le \omega t \le \pi \qquad \text{(B)}$$

During negative half cycle of v_i, D_2 conducts and D_1 is off. i_o and v_o are still given by Equations (A) and (B) respectively. The waveforms of i_o and v_o are shown below.





(b) v_o is a full rectified voltage with peak value of 100 V

$$\therefore \quad V_{dc} = 0.636\,[100 \text{ V}]$$

$$= 63.6 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R}$$

$$= \frac{63.6 \text{ V}}{10 \text{ k}\Omega} = 6.36 \text{ mA}$$

(c) $$i_{d_1} = i_o + i_1 \qquad \text{[From Fig. B]}$$

$$i_1 = \frac{v_i}{R_1} = \frac{200 \sin \omega t\ V}{10 \text{ k}\Omega} = 20 \sin \omega t \text{ mA}$$

$$\therefore \quad i_{d_1} = 10 \sin \omega t + 20 \sin \omega t$$

$$= 30 \sin \omega t \text{ mA}, \qquad 0 \le \omega t \le \pi$$

During the negative half cycle of v_i, D_2 is on and D_1 is off. The diode current is a half-rectified sinusoid with a peak value of 30 mA.

∴ Peak diode current is

$$I_{dm} = 30 \text{ mA}$$

$$I_{dc\ (diode)} = \frac{I_{dm}}{\pi} = \frac{30 \text{ mA}}{\pi} = 9.55 \text{ mA}$$

(d) To find PIV let us consider a part of the circuit of Fig. (A) which is shown in Fig. C.

KVL equation to the circuit of Fig. (C) is

$$v_d - v_o = 0$$
$$v_d = v_o$$
$$\text{PIV} = v_{omax}$$
$$= 100 \text{ V}$$

PIV rating of diode must be greater than 100 V.

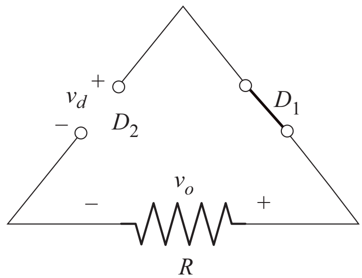


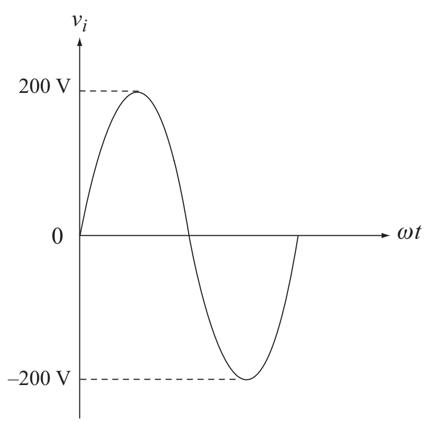
Fig. C

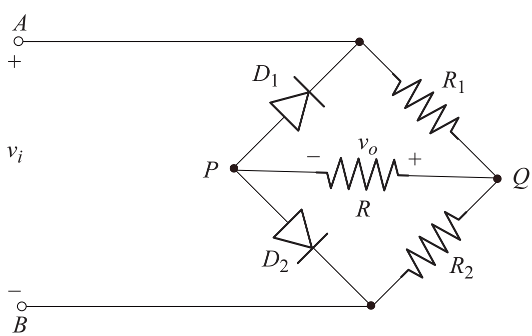
Example 1.29

For the circuit shown below

(a) Explain the operation
(b) Calculate average ouput voltage and current
(c) Calculate average and peak diode current
(d) PIV across each diode.

Assume ideal diodes.
Take $R_1 = R_2 = R = 10$ kΩ

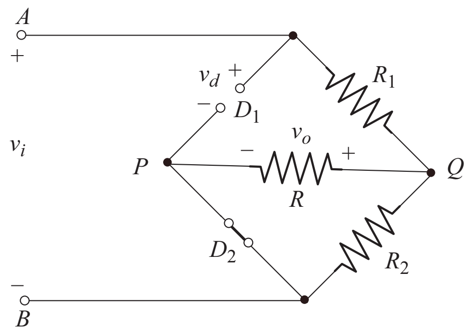




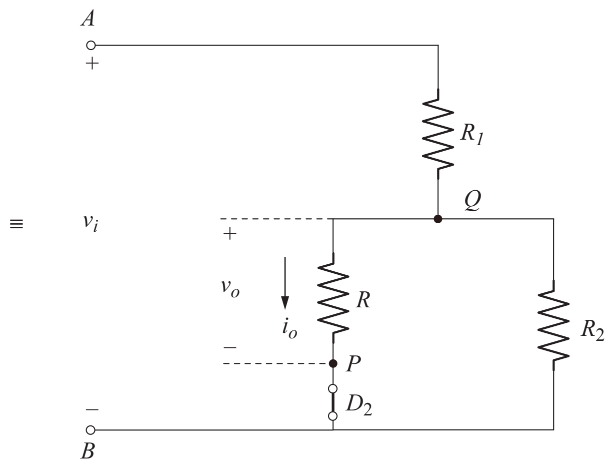
Solution

(a) Circuit Operaton

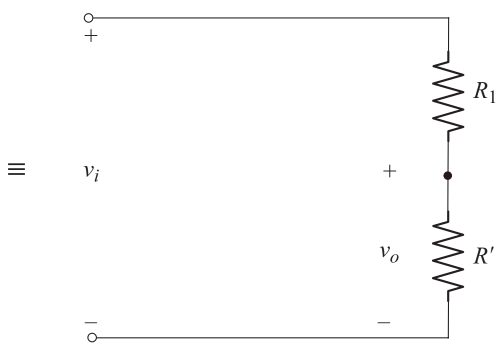
During the positive half cycle of v_i, potential of point A is positive with respect to point B. D_2 conducts and D_1 is reverse biased. The equivalent circuit is shown below.



(a)



(b)



(c)

Let
$$R' = R \,||\, R_2$$
$$R' = 10\text{ k}\Omega \,||\, 10\text{ k}\Omega = 5\text{ k}\Omega$$

Using voltage division rule in the circuit of Fig. (c), we have

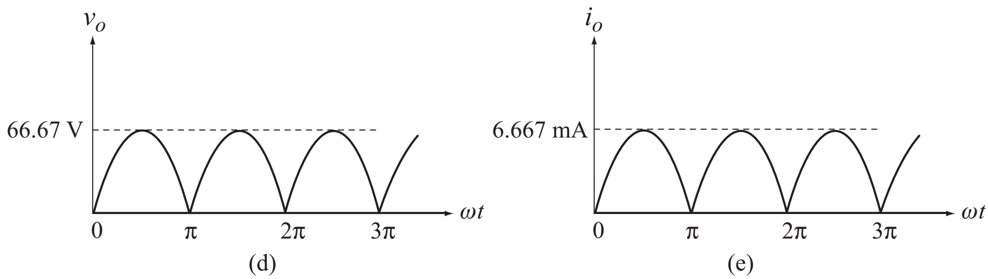
$$v_o = \frac{v_i\, R'}{R' + R_1}$$
$$= 200 \sin \omega t \left[\frac{5\text{ k}\Omega}{10\text{ k}\Omega + 5\text{ k}\Omega}\right]$$
$$v_o = 66.67 \sin \omega t \text{ V}, \quad 0 \le \omega t \le \pi \qquad \text{(A)}$$

$$i_o = \frac{v_o}{R}$$

$$= \frac{66.67 \sin \omega t \text{ V}}{10 \text{ k}\Omega}$$

$$i_o = 6.667 \sin \omega t \text{ mA}, \quad 0 \leq \omega t \leq \pi \qquad \text{(B)}$$

During the negative half cycle of v_i, the roles of D_1 and D_2 are interchanged. v_o and i_o are still given by Equations (A) and (B) respectively. The waveforms of v_o and i_o are shown below.



Note that v_o is a full rectified voltage.

From the waveforms shown above, we note that

Peak output voltage = 66.67 V

Peak output current = 6.667 mA

(b) dc output voltage is

$$V_{dc} = 0.636\,[66.67 \text{ V}]$$

$$= 42.4 \text{ V}$$

dc output current is

$$I_{dc} = \frac{V_{dc}}{R} = \frac{42.4 \text{ V}}{10 \text{ k}\Omega} = 4.24 \text{ mA}$$

(c) Average diode current is

$$I_{dc\ (diode)} = \frac{I_{dc}}{2} = \frac{4.24 \text{ mA}}{2} = 2.12 \text{ mA}$$

Peak diode current is same as peak output current

$$\therefore \qquad I_{dm} = I_{om} = 6.667 \text{ mA}$$

(a) To find PIV let us apply KVL to the path $A - D_1 - P - D_2 - B - A$ in the circuit of Fig. (a).

$$v_i - v_d = 0$$

$$v_d = v_i$$

$$\text{PIV} = v_{i\max} = 200 \text{ V}$$

1.16 PRACTICAL RECTIFIER CIRCUITS

Most of the electronic systems require dc voltages in the range of 5 V – 30 V for their proper operation. Since the available $1-\phi$ ac supply is 220 V at 50 Hz, the ac voltage is first reduced in amplitude using a step down transformer and then fed to the rectifier. The circuits of half-wave and full wave bridge rectifier using a step down transformer is shown in Fig. 1.25.

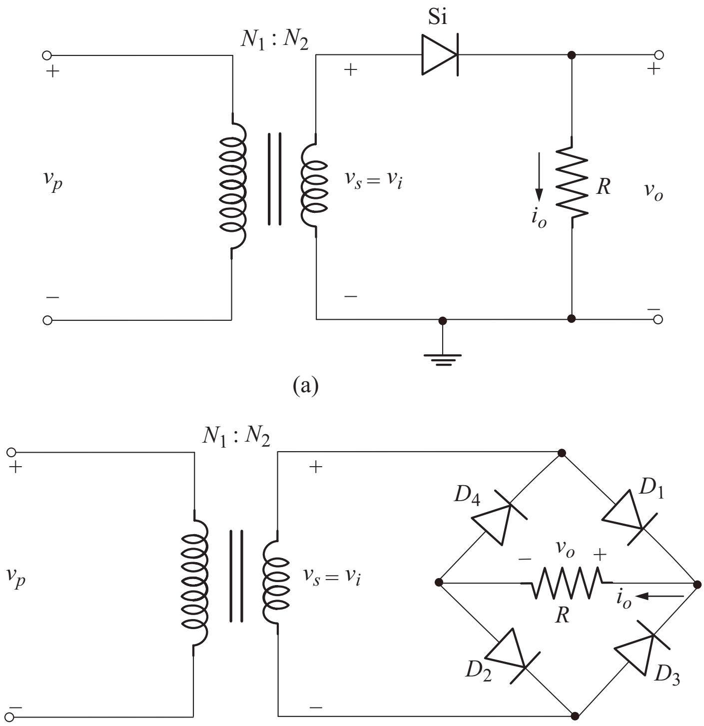


Fig. 1.25 (a) Half wave rectifier
(b) Full wave bridge rectifier

Let v_P = instantaneous primary voltage

v_S = instantaneous secondary voltage

$$\frac{N_1}{N_2} = \text{turns ratio of the transformer}$$

$$\frac{v_P}{v_S} = \frac{N_1}{N_2} \tag{1.50}$$

It should be noted that v_S is same as v_i which is the actual voltage applied to the rectifier. The following two examples illustrates how we can proceed to solve for various currents and voltages when the value of v_p is known

Let $v_p = 311 \sin \omega t$ V

$$\text{and} \qquad \frac{N_1}{N_2} = 10 \text{ or } 10:1$$

$$\text{Now} \qquad v_S = v_i = \frac{N_2}{N_1} v_p$$

$$= \frac{311}{10} \sin \omega t$$

$$v_i = 31.1 \sin \omega t = V_m \sin \omega t$$

$$\Rightarrow \qquad V_m = 31.1 \text{ V}$$

This value of V_m should be used to calculate the required currents and voltages as illustrated in the previous examples. As another example

$$\text{Let} \qquad V_p = 220 \text{ V (rms)}$$

$$\text{and} \qquad \frac{N_1}{N_2} = 10 \text{ or } 10:1$$

$$\text{Now} \qquad V_s = V_i = \frac{N_2}{N_1} V_p$$

$$= \frac{220 \text{ V}}{10}$$

$$= 22 \text{ V (rms)}$$

$$V_m = \sqrt{2} \times \text{RMS value}$$

$$= \left(\sqrt{2}\right) (22 \text{ V})$$

$$= 31.11 \text{ V}$$

1.17 FULL WAVE RECTIFIER USING TWO DIODES AND A CENTRE-TAPPED TRANSFORMER

A full wave rectifier can also be constructed using only two diodes but it requires a centre–tapped transformer as shown in Fig. 1.26. The centre tapped transformer is used to obtain two equal voltages but of opposite phase at points *A* and *B*.

Note that the secondary has two identical windings and each half of the secondary having N_2 turns along with the primary having N_1 turns works as a transformer with turns ratio

$$\frac{N_1}{N_2} = \frac{v_p}{v_S} = \frac{v_p}{v_i} \tag{1.51}$$

The voltage from one end i.e., *A* (or *B*) to centre-tap i.e., *O* is v_S and the end-to-end (between *A* and *B*) voltage is $2\,v_S$.

During the positive half cycle of v_p, the voltage at point *A* is positive going and that at *B* is negative going with respect to centre tap. D_1 conducts and D_2 is off. The current follows the path $A - D_1 - \text{P} - O - A$. The equivalent circuit is shown in Fig. 1.27 (a).

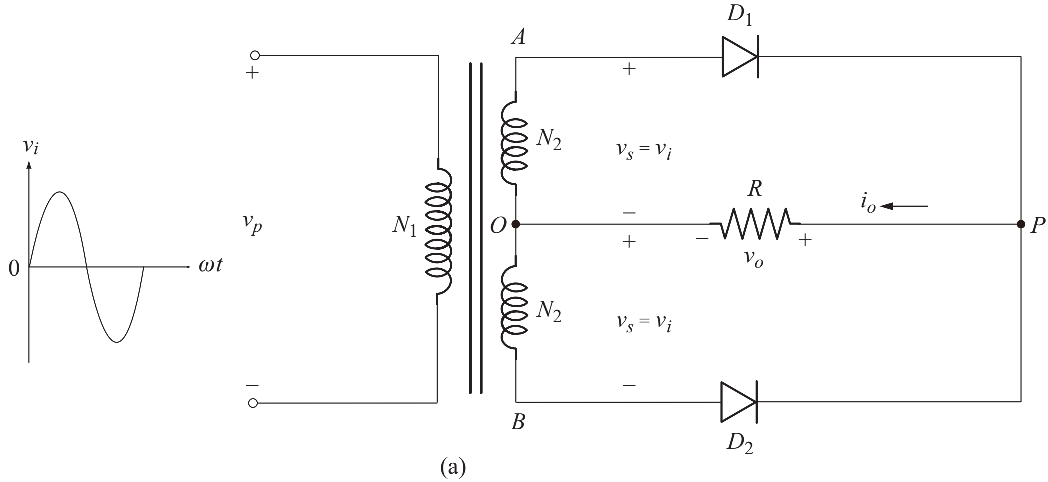


Fig. 1.26 Full wave rectifier with centre-tapped transformer

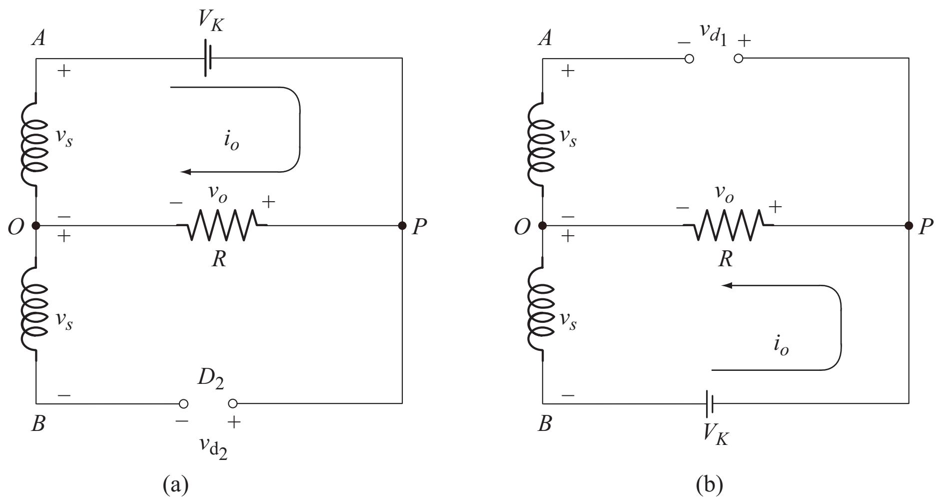


Fig. 1.27 Equivalent circuit
(a) During the positive half cycle of input
(b) During the negative half cycle of input

Applying KVL to the path $A - P - O - A$ in the circuit of Fig. 1.27(a) we have

$$-V_K - v_o + v_s = 0$$

$$v_o = v_s - V_K$$

When $v_s = V_m$, the peak level of the output voltage is

$$V_{om} = V_m - V_K$$

Therefore the equation for output voltage is

$$\begin{aligned} v_o &= V_{om} \sin \omega t \\ &= (V_m - V_K) \sin \omega t, \quad 0 \le \omega t \le \pi \end{aligned} \tag{1.52}$$

The output current is

$$i_o = \frac{v_o}{R}$$

$$i_o = \frac{V_m - V_K}{R} \sin \omega t \quad 0 \le \omega t \le \pi \tag{1.53}$$

During the negative half cycle of v_p, the voltage at point A is negative going and that at B is positive going with respect to the centre tap. D_2 conducts and D_1 is off. The current follows the path $B - D_2 - P - O - B$. The equivalent circuit is shown in Fig. 1.27(b). Note that during both half cycles of ac input, the load current i_o flows from P to O only. Hence i_o is unidirectional or dc. The waveforms of v_o and i_o are shown in Fig. 1.28.

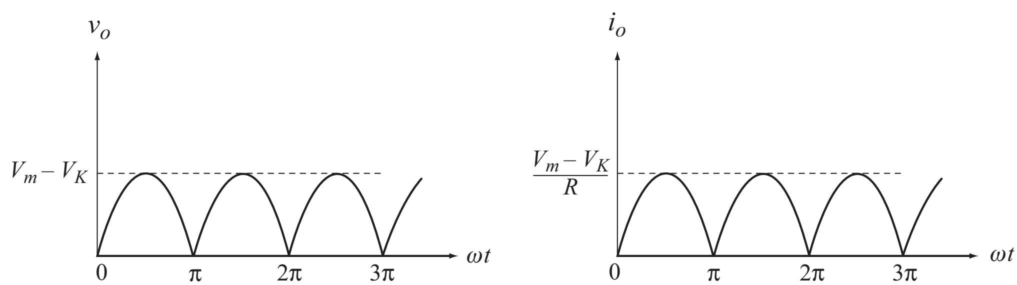


Fig. 1.28 Wave forms of v_o and i_o

The expressions for average output voltage, average output current etc can be derived in a similar way as explained in section 1.15 for full-wave bridge rectifier.

The average output voltage is

$$V_{dc} = 0.636\, [V_m - V_K] \tag{1.54}$$

Average output current is

$$I_{dc} = 0.636\, \frac{[V_m - V_K]}{R} \tag{1.55}$$

Average diode current is

$$\begin{aligned} I_{dc\ (diode)} &= \frac{I_{dc}}{2} \\ &= 0.318\, \frac{[V_m - V_K]}{R} \end{aligned} \tag{1.56}$$

The peak diode current is

$$I_{dm} = I_{om} = \frac{V_m - V_K}{R} \tag{1.57}$$

and the diode power dissipation is

$$P_D = V_K I_{dc\ (diode)} \tag{1.58}$$

***Peak Inverse Voltage* [*PIV*]**

Applying KVL to the path $A - P - B - O - A$ to the circuit of Fig. 1.27 (a) we have

$$- V_K - v_{d_2} + v_s + v_s = 0$$

$$v_{d_2} = 2\, v_s - V_K$$

The reverse voltage v_{d_2} is maximum when $v_s = V_m$

$$\therefore \qquad \text{PIV} = 2\, V_m - V_K$$

Neglecting V_K, we get

$$\text{PIV} \approx 2\, V_m \tag{1.59}$$

Note that the PIV across each diode is twice that of the bridge circuit. For safe operation, the PIV rating of the diode must be greater than $2V_m$.

Example 1.30

A full-wave rectifier using two diodes and a centre-tapped transformer is supplying a resistive load of 2 kΩ. The ac supply voltage is 220 V (rms) and turns ratio of the transformer is 10 : 1. Assuming silicon diodes calculate

(a) dc output voltage
(b) dc load current
(c) dc diode current
(d) Peak diode current
(e) Peak load current
(f) PIV across each diode

Solution

$$V_p = 220 \text{ V (rms)} \qquad R = 2 \text{ k}\Omega$$

$$\frac{N_1}{N_2} = 10$$

$$V_s = \frac{N_2}{N_1} V_p = \frac{1}{10} (220 \text{ V}) = 22 \text{ V}$$

$$V_m = \sqrt{2}\ V_s = \sqrt{2}\ (22 \text{ V}) = 31.11 \text{ V}$$

(a) dc output voltage is

$$V_{dc} = 0.636\,[V_m - V_K]$$
$$= 0.636\,[31.11\text{ V} - 0.7\text{ V}] = 19.34\text{ V}$$

(b) dc load current is

$$I_{dc} = \frac{V_{dc}}{R} = \frac{19.34\text{ V}}{2\text{ k}\Omega} = 9.67\text{ mA}$$

(c) dc diode current is

$$I_{dc\ (diode)} = \frac{I_{dc}}{2} = 4.835\text{ mA}$$

(d) peak diode current is

$$I_{dm} = \frac{V_m - V_K}{R} = \frac{31.11\text{ V} - 0.7\text{ V}}{2\text{ k}\Omega} = 15.2\text{ mA}$$

(e) peak load current is same as peak diode current

$$I_{om} = 15.2\text{ mA}$$

(f) PIV across each diode is

$$\text{PIV} = 2V_m = 2\,(31.11\text{ V}) = 62.22\text{ V}$$

◆ 1.18 ANALYSIS OF RECTIFIER CIRCUITS CONSIDERING THE EFFECT OF DIODE RESISTANCE

In the analysis carried out so far, we have not taken into account the average forward resistance r_{av} that appears in the diode equivalent circuit, since we have assumed that, $R \gg r_{av}$. If r_{av} is comparable to R, then it is necessary to consider the effect of r_{av}. The average forward resistance r_{av}, is some times also denoted by r_f. In the following example we have considered the effect of r_{av}.

Example 1.31

Repeat example 1.30 with $R = 100\ \Omega$ and $r_{av} = 20\ \Omega$.

Solution

Here we first find the peak diode / load current

$$I_{dm} = I_{om} = \frac{V_m - V_K}{R + r_{av}}$$

$$= \frac{31.11\text{ V} - 0.7\text{ V}}{100\ \Omega + 20\ \Omega} = 253.42\text{ mA}$$

Average load current is

$$I_{dc} = \frac{2}{\pi} I_{om} = 0.636\,[253.42\text{ mA}] = 161.17\text{ mA}$$

Average diode current is

$$I_{dc\,(diode)} = \frac{I_{dc}}{2} = 80.58 \text{ mA}$$

PIV across each diode is

$$\text{PIV} = 2V_m = 2\,[31.11 \text{ V}] = 62.22 \text{ V}$$

Average load voltage is

$$V_{dc} = I_{dc} R = (161.17 \text{ mA})\,(100\ \Omega) = 16.11 \text{ V}$$

Note:

- For full wave bridge rectifier

$$I_{om} = I_{dm} = \frac{V_m - 2\,V_K}{R + 2\,r_{av}}$$

- For half wave rectifier

$$I_{om} = I_{dm} = \frac{V_m - V_K}{R + r_{av}}$$

◆ 1.19 CLIPPING CIRCUITS OR LIMITING CIRCUITS

Clipping circuits (clippers) are used to remove a portion of a time varying input signal without distorting the remaining part of the applied waveform. One simple example of a clipper is a half–wave rectifier which transfers only one half cycle of the input to the output, while clipping-off the other half cycle. Therefore diodes can be used to perform clipping.

◆ 1.20 SHUNT OR PARALLEL CLIPPER

In parallel clipper the diode appears in the parallel branch or shunt with the applied input signal. Figure 1.29 shows a shunt clipper with a bias or reference voltage V_R.

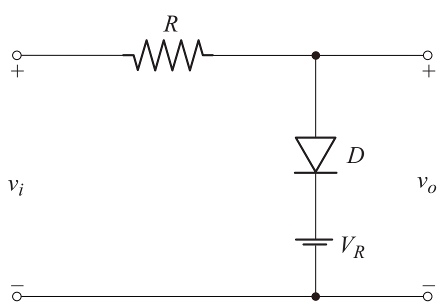


Fig. 1.29 Shunt clipper

The input signal v_i can be any periodic signal such as sine, square, triangle etc with peak value V_m greater than V_R. If $V_m < V_R$, clipping does not take place. In the analysis of clipping circuits we use the approximate equivalent circuit of diode, unless otherwise specifically mentioned.

Just before the diode conducts, the current through R is zero and hence the input signal v_i is directly available at the anode of diode, as shown in Fig. 1.30.

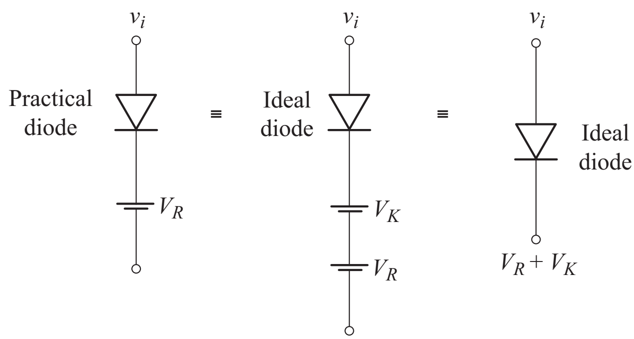


Fig. 1.30 Voltage at diode terminals

For the ideal diode to conduct, it is enough that the anode voltage just equals the cathode voltage. From Fig. 1.30 we find that the diode conducts for

$$v_i \geq V_R + V_K$$

The equivalent circuit during diode conduction is shown in Fig. 1.31 (a).

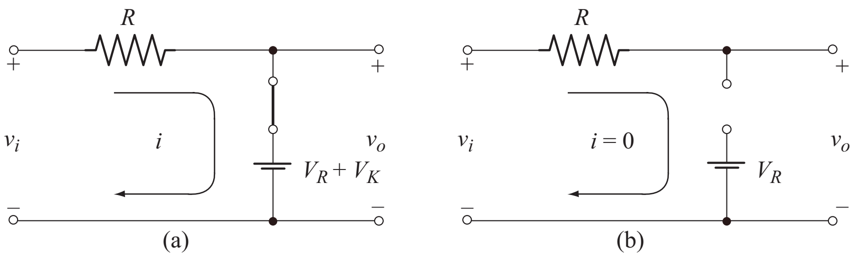


Fig. 1.31 Equivalent circuit during
(a) Diode conduction
(b) Diode off

From the equivalent circuit of Fig. 1.31 (a), we find that

$$v_o = V_R + V_K \quad \text{for} \quad v_i \geq V_R + V_K \tag{1.60}$$

For $v_i < V_R + V_K$, the diode is off and the equivalent circuit is shown in Fig. 1.31 (b). Writing KVL equation to the equivalent circuit we have

$$v_i - iR - v_o = 0$$

or

$$v_o = v_i, \quad \text{for} \quad v_i < V_R + V_K \tag{1.61}$$

In Fig. 1.32, the ouput waveforms are sketched for sine, square and triangular input waveforms.

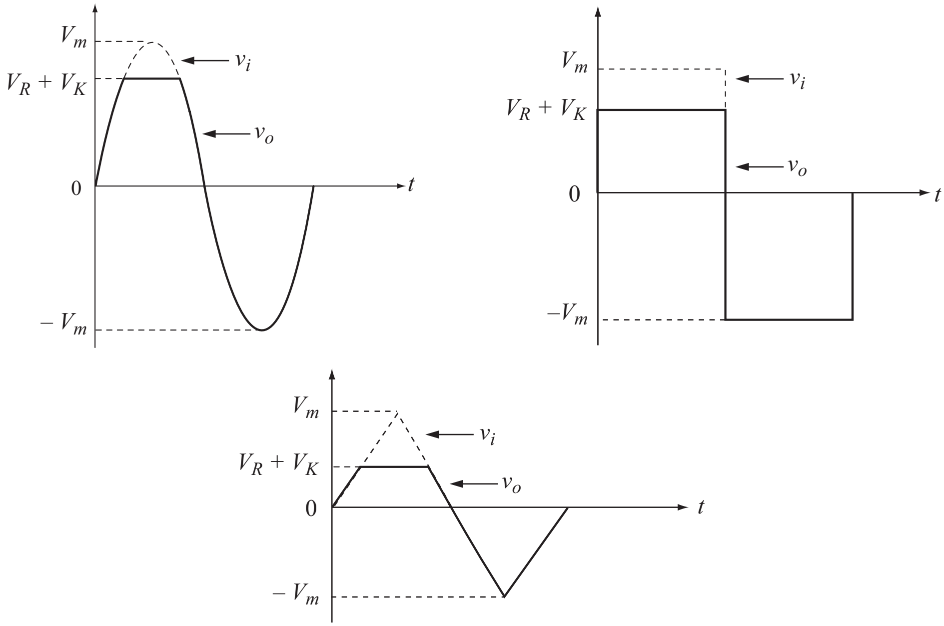


Fig. 1.32 Output waveforms of series clipper of Fig. 1.29 for different inputs

Note that the circuit clips-off a portion of the input signal, which lies above the level $V_R + V_K$ and retains the remaining part as it is.

Transfer Characteristics

The transfer characteristics is obtained by plotting v_o as a function of v_i. Transfer characteristics can be easily constructed by evaluating slope $\dfrac{\Delta v_o}{\Delta v_i}$

For $v_i \geq V_R + V_K$, from Equation (1.60) we have

$$v_o = V_R + V_K = \text{constant}$$

$$\Rightarrow \qquad \Delta v_o = 0$$

$$\text{Hence} \qquad \text{slope} = \frac{\Delta v_o}{\Delta v_i} = 0 \tag{1.62}$$

For $v_i < V_R + V_K$, from Equation (1.61) we have

$$v_o = v_i$$

$$\Rightarrow \qquad \Delta v_o = \Delta v_i$$

$$\therefore \qquad \text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 1 \tag{1.63}$$

Figure 1.33 shows the transfer characteristics of the shunt clipper shown in Fig. 1.29 along with output waveform for sinusoidal input.

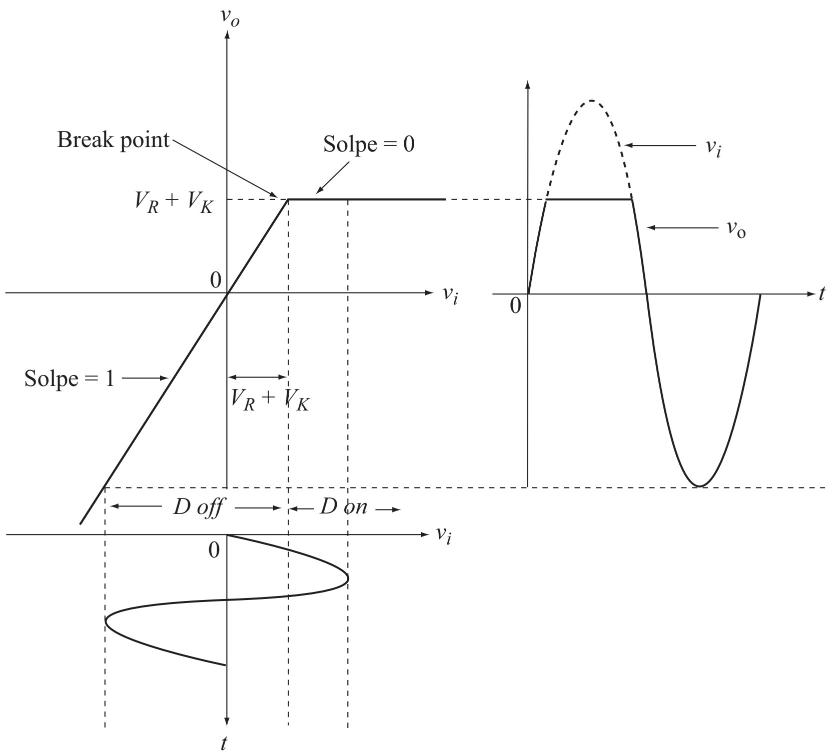


Fig. 1.33 Transfer characteristics and output waveform for the shunt clipper of Fig. 1.29

◆ 1.21 SERIES CLIPPER

In series clipper, the diode appears in series with the input or it appears in the series branch as shown in Fig. 1.34.

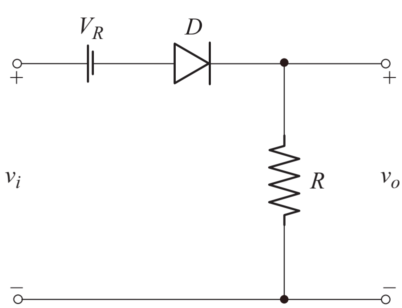


Fig. 1.34 Series clipper

First let us find the voltage on diode terminals as shown in Fig. 1.35.

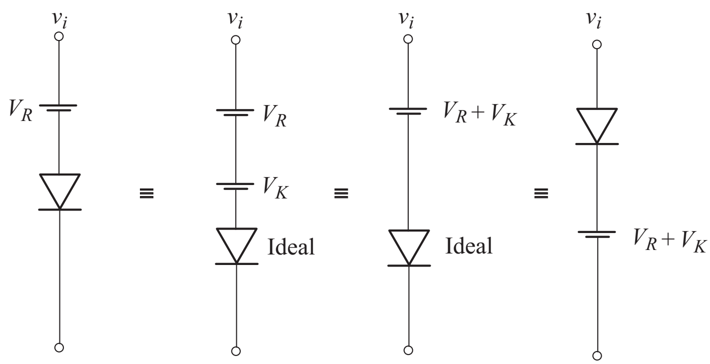


Fig. 1.35 voltage on diode terminals

From Fig. 1.35 we note that, diode conducts for $v_i \geq V_R + V_K$. The equivalent circuit during diode conduction is shown in Fig. 1.36 (a).

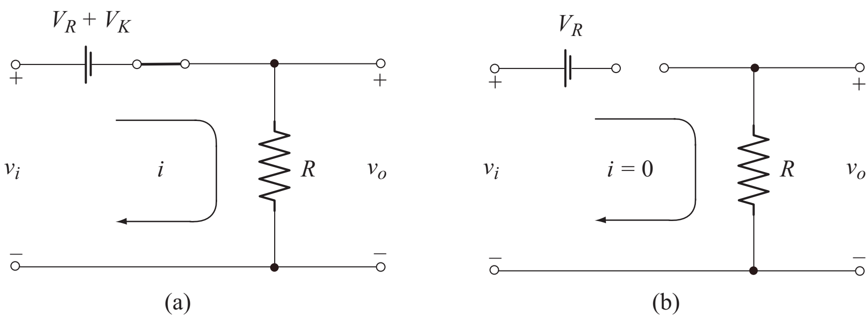


Fig. 1.36 Equivalent circuit
(a) During diode conduction
(b) During diode off

Applying KVL to the circuit of Fig. 1.36(a) we have

$$v_i - [V_R + V_K] - v_o = 0$$

$$v_o = v_i - [V_R + V_K], \quad \text{for } v_i \geq V_R + V_K \tag{1.64}$$

Since $V_R + V_K$ is a constant

$$\Delta v_o = \Delta v_i$$

$$\Rightarrow \qquad \text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 1$$

Alo from Equation (1.64), when $v_i = V_m$,

$$v_o = V_m - [V_R + V_K] \tag{1.65}$$

For $v_i < V_R + V_K$, diode is off. The equivalent circuit is shown in Fig. 1.36(b). From the equivalent circuit we have

$$v_o = iR = 0 \tag{1.66}$$

Also $$\text{slope} = \frac{\Delta v_o}{\Delta v_i} = 0 \tag{1.67}$$

The transfer characteristics and the output voltage waveforms for different inputs are shown in Fig. 1.37.

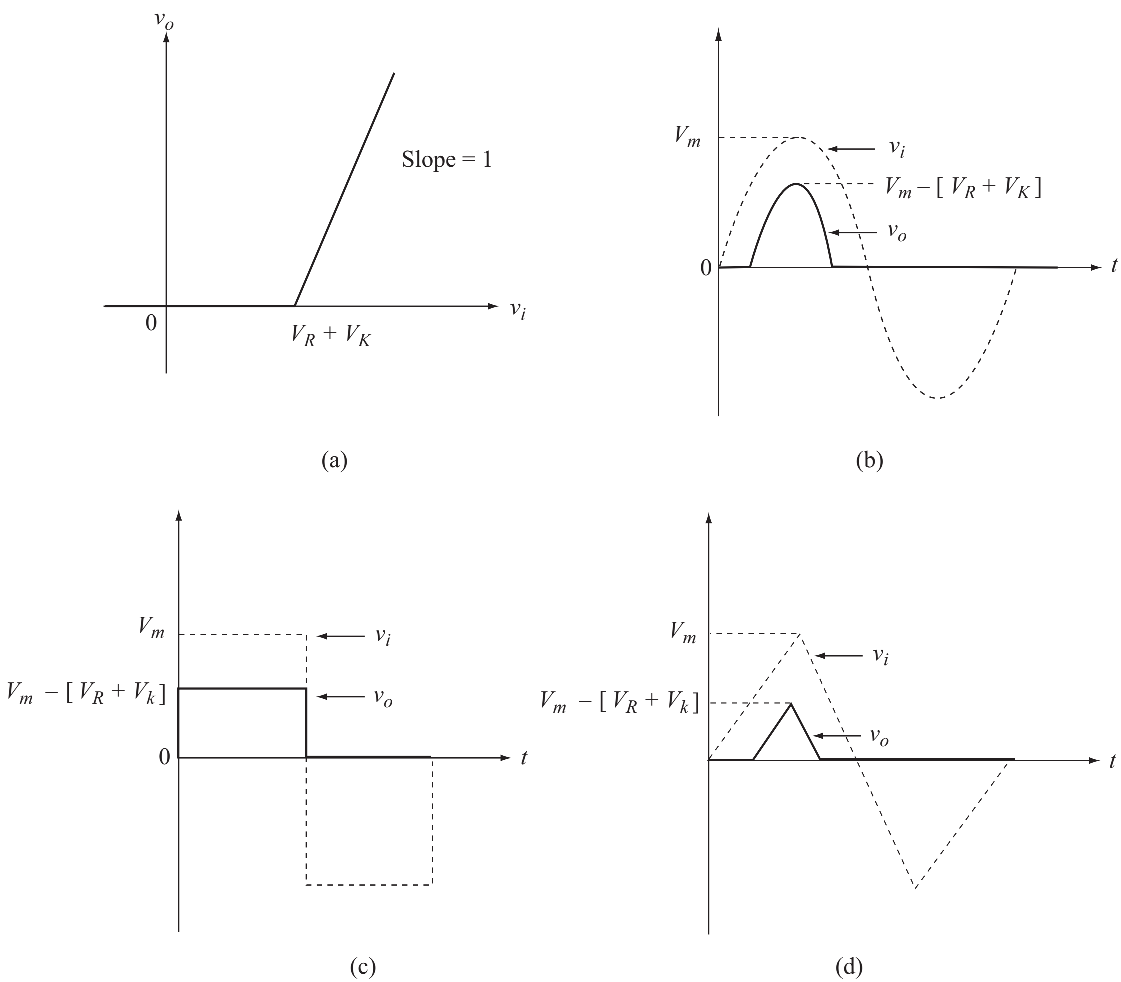


Fig. 1.37 (a) Transfer characteristics
(b) (c) and (d) output waveforms sine, square and triangular inputs

Note: In the clipping circuits, the reference voltage V_R need not be in series with the diode. The location of diode and reference voltage depends on the nature of the required output waveform.

◆ 1.21 CLIPPING AT TWO INDEPENDENT LEVELS [DOUBLE ENDED CLIPPER]

In Section 1.20 we saw clipper which clipped at one reference level. Two such clippers can be combined to obtain clipping at two independent levels. Fig. 1.38 shows clipping circuit which uses two reference voltages.

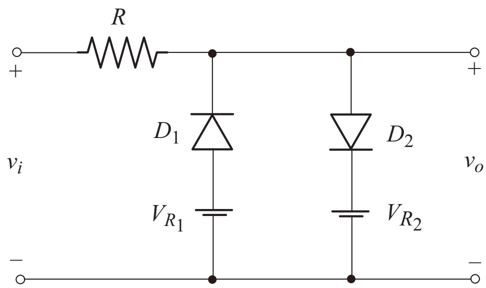


Fig. 1.38 Clipping circuit with two reference voltages

Note that both V_{R_1} and V_{R_2} are positive. V_{R_1} forward biases D_1 and V_{R_2} reverse biases D_2. Also $V_{R_2} > V_{R_1}$.

Now let us find the voltages on the terminals of D_1 as shown in Fig. 1.39.

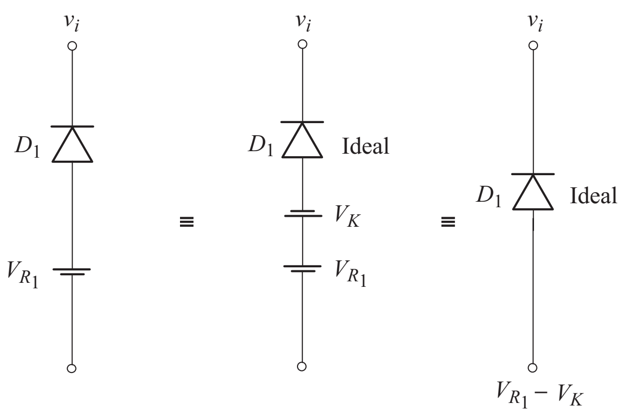


Fig. 1.39 Terminal voltages on D_1

From Fig. 1.39, we note that D_1 conducts for $v_i \le V_{R_1} - V_K$.

The equivalent circuit is shown in Fig. 1.40 (a). From the equivalent circuit, we find that

$$v_o = V_{R_1} - V_K, \quad \text{for } v_i \le V_{R_1} - V_K \tag{1.68}$$

$$\text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 0 \tag{1.69}$$

The analysis of the second branch containing D_2 and V_{R_2} has already been considered in section 1.20. We find that D_2 conducts for $v_i \geq V_{R2} + V_K$.

The equivalent circuit is shown in Fig. 1.40(c). From the equivalent circuit we find that

$$v_o = V_{R_2} + V_K, \quad \text{for } v_i \geq V_{R_2} + V_K \tag{1.70}$$

$$\text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 0 \tag{1.71}$$

For $(V_{R_1} - V_K) < v_i < (V_{R_2} + V_K)$, neither D_1 nor D_2 conducts.

The equivalent circuit is shown in Fig. 1.40(b). From the equivalent circuit we get

$$v_o = v_i \tag{1.72}$$

$$\text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 1 \tag{1.73}$$

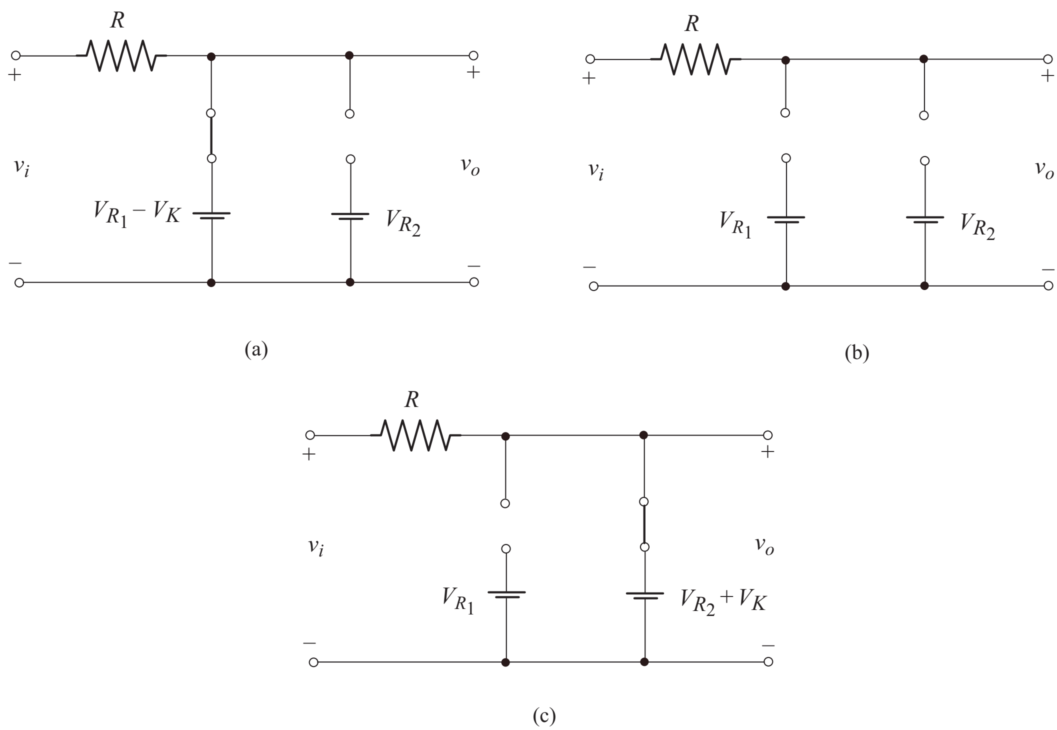


Fig. 1.40 Equivalent circuit for
(a) $v_i \leq V_{R_1} - V_K$
(b) $V_{R_1} - V_K < v_i < V_{R_2} + V_K$
(c) $v_i \geq V_{R_2} + V_K$

The circuit operation is summarized in Table 1.4.

Table 1.4 Summary of operation of double ended clipper of Fig. 1.38

Input voltage	*Diode status*	*Output voltage*	*Slope*
$v_i \leq V_{R_1} - V_K$	D_1 on D_2 off	$v_o = V_{R_1} - V_K$	0
$V_{R_1} - V_K < v_i < V_{R_2} + V_K$	D_1 off D_2 off	$v_o = v_i$	1
$v_i \geq V_{R_2} + V_K$	D_1 off D_2 on	$v_o = V_{R_2} + V_K$	0

The transfer characteristic along with output waveform for sinusoidal input is shown in Fig. 1.41.

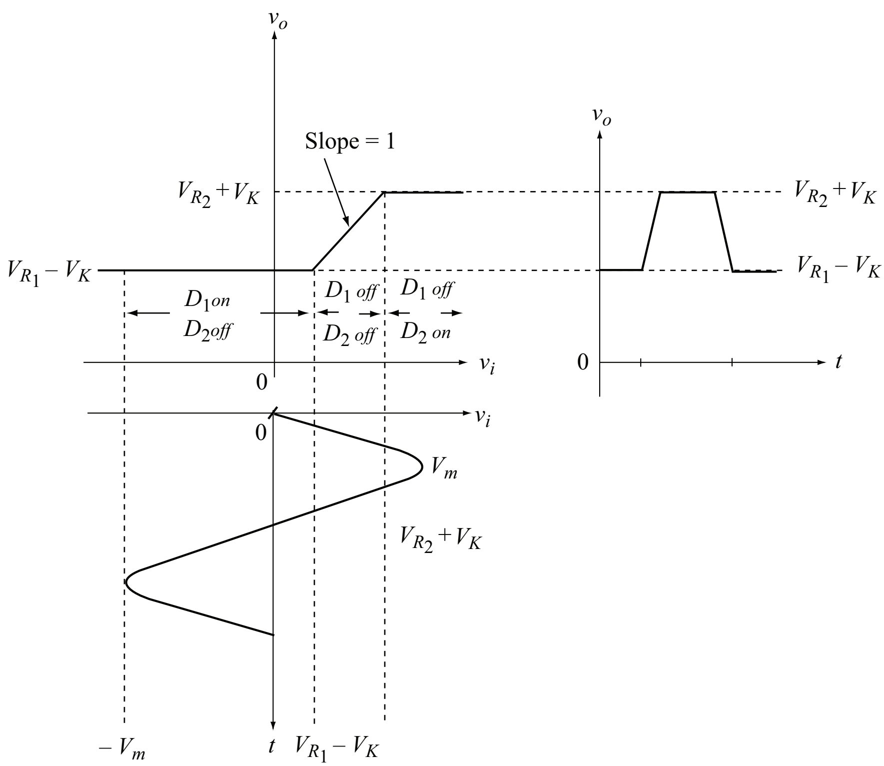


Fig. 1.41 Transfer characteristic and output waveform of the double ended clipper of Fig. 1.38

The output voltage waveforms for other input signals is shown in Fig. 1.42.

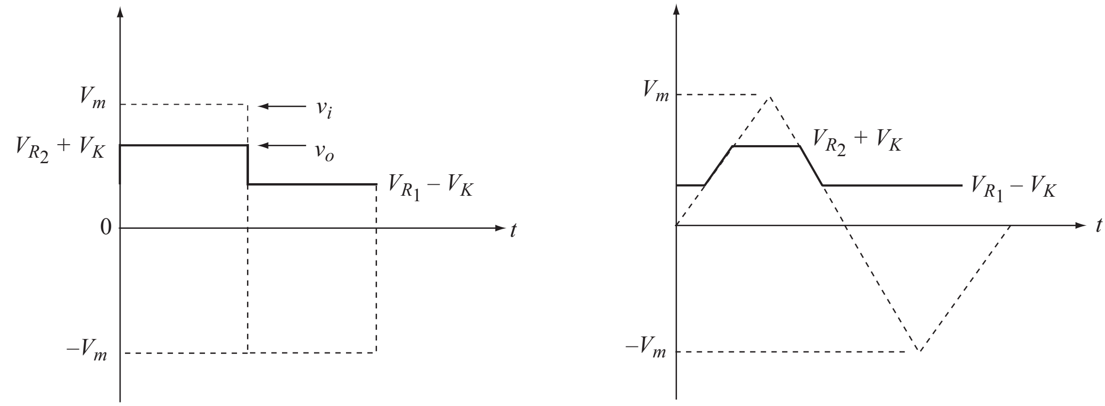


Fig. 1.42 Output waveforms for square and triangular input waveforms

◆ 1.22 ANOTHER DOUBLE ENDED CLIPPER

The variation in the double ended clipper results as shown in Fig. 1.43 when the polarity of V_{R_1} is reversed in the circuit of Fig. 1.38.

Since V_{R_1} is negative, the condition for D_1 to conduct becomes

$$v_i \leq -[V_{R_1} + V_K]$$

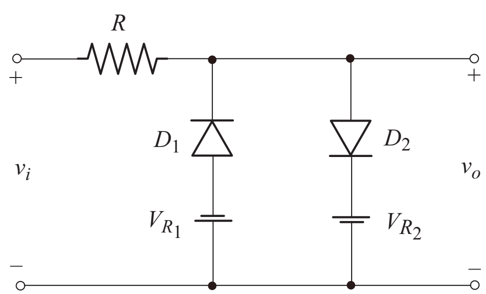


Fig. 1.43 Another double ended clipper

The analysis of this circuit can be carried out in a similar way as explained in the previous section. The circuit operation is summarized in Table 1.5.

Table 1.5 Summary of operation of doubled ended clipper of Fig. 1.43

Input voltage	*Diode status*	*Output voltage*	*Slope*
$v_i \leq -[V_{R_1} + V_K]$	D_1 on D_2 off	$v_o = -[V_{R_1} + V_K]$	0
$-[V_{R_1} + V_K] < v_i < [V_{R_2} + V_K]$	D_1 off D_2 off	$v_o = v_i$	1
$v_i \geq V_{R_2} + V_K$	D_1 off D_2 on	$v_o = [V_{R_2} + V_K]$	0